Novel Harmonic-Based Scheme for Mapping Rice-Crop Intensity at a Large Scale Using Time-Series Sentinel-1 and ERA5-Land Datasets

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***Abstract*— Rice-crop intensity is the annual number of rice growth cycles in a field. Monitoring the intensity on a large scale is vital in evaluating grain production and its ecological impact. Synthetic aperture radar (SAR) has an all-weather imaging capability. However, the existing SAR-based rice-crop intensity mapping methods mostly focus on small regions due to the diversity of rice backscatter patterns, the inefficiency of the time-series feature extraction, and the unavailability of rice phenological information on a large scale. In this study, a harmonic-based method is proposed to identify the essential backscatter periodicities. It also suppresses short-term disturbance in time-series Sentinel-1 SAR data without setting filtering windows or assuming profile shapes. The method detects backscatter troughs, eliminating the requirement for point-by-point traversal mathematical operations. Annual temperature profiles are derived from time-series ERA5-Land data to identify troughs related to rice growth cycles under various agro-climatic conditions. Then, the single (135 537 km^2), double (19 036 km^2), and triple (259 km^2) rice-crop intensities covering the entire Southern China in 2020 are mapped in a 10-m resolution, without relying on region-specific prior phenological information. The method achieves an overall accuracy of 82.26% and can potentially support the continental or global mapping task.**

***Index Terms*— Harmonic analysis, rice-crop intensity, synthetic aperture radar (SAR), temperature.**

I. Introduction

A RICE-CROP intensity is the number of rice growth cycles per year in the same field, implying the temporal replications of the rice planting area. Increasing rice-crop intensity is a common practice for increasing rice yield in hot and humid areas, where farmers can cultivate up to three cycles annually [1], [2]. However, rice cropping consumes a lot of water resources, producing greenhouse gases [3], such as methane and carbon dioxide. As a result, increasing rice-crop intensity also adversely impacts agricultural sustainability (e.g., soil fertility) and ecosystem health (e.g., biodiversity and carbon balance). Monitoring rice-crop intensity at a large scale can provide recommendations for achieving a harmonious balance between agricultural expansion and ecological preservation.

Optical remote sensing data are traditionally the preferred choice for mapping rice-crop intensity because various optical indices can be obtained for identifying the phenological changes [4], [5]. However, frequent foggy and cloudy weather, attributed to the subtropical monsoon climate in Southern China, results in severe observation missingness. Although utilizing the interpolation algorithm for cloud gaps or the synergistic integration of multiple optical satellite datasets can partially impute the missingness, constructing year-long time-series observations free from cloud interference at a large scale remains challenging. Furthermore, relative radiometric calibration among different satellites may introduce additional noise sources. The existing rice-crop intensity mapping covering the entire Southern China primarily relies on Moderate Resolution Imaging Spectroradiometer (MODIS) due to its short revisit cycle (one or two days), which allows for the maximum possible number of cloud-free observations during the rice growing season [6], [7], [8]. However, MODIS datasets have a moderate spatial-resolution ($\geq$250 m). Due to uneven terrain in Southern China, rice paddies for individual farmers are typically small, ranging from 10 $\times$ 10 to 100 $\times$ 100 m, and are often adjacent to other land cover types. The rice growth process monitored with the moderate-resolution data is concerned.

The operational spaceborne synthetic aperture radar (SAR) sensor of the Sentinel-1 mission provides high spatiotemporal resolution data under all weather conditions [9], [10]. The vertically transmit and horizontally receive (VH) backscatter is sensitive to the changes in moisture and roughness of the rice canopy and underlying surface [11], [12]. Thus, rice-crop intensity can be assessed if critical phenological periods are identifiable in the time-series VH backscatter. Some studies have managed to map the rice-crop intensity on

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a small scale [13], [14], [15]. However, three critical issues significantly limit the accuracy of the large-scale mapping based on the time-series SAR data, including the diversity of rice backscatter patterns, the inefficiency of time-series feature extraction, and the unavailability of rice phenological information.

The existing research generally suggests that rice backscatter reaches a temporal trough in the transplanting period and a crest in the reproductive period [16], [17], [18]. However, rice backscatter patterns exhibit a wide range of variations due to complex cultivation practices, including different rice varieties, planting densities, field management techniques (e.g., irrigation, drainage, and fertilization), and climate conditions [2], [19]. Cultivation practices notably impact backscatter from rice fields because they influence the dielectric and geometric properties of the rice canopy and the underlying surface. This influence can occasionally change the timing, quantity, and backscatter intensity of the troughs and crests, posing challenges in establishing a representative rice backscatter pattern. Thus, reliable backscatter features are demanded to identify a rice growth cycle accurately.

Since time-series backscatter can be significantly influenced by noise in SAR data, irregular (amount and timing) rainfalls, and human-induced disturbances, temporal filtering is necessary before conducting phenological analysis [18], [19], [20]. However, the window size affects the commonly used windowed statistical methods (e.g., the moving average [21] and maximum value composite [22] ones), while the local fitting methods (e.g., the Savitzky–Golay [23], logistic [24], and Gaussian [25] fitting ones) are restricted to recognizing specific profile shapes, thereby limiting their ability to capture the overall trend of diverse time series. Moreover, identifying rice phenology involves detecting troughs or crests in the time-series profiles, which traditionally requires point-by-point differentiation and comparison. The operations can be time-consuming and computationally intensive for national-scale analysis. Complex rules may further be required to validate the detected features [26], [27]. Harmonic decomposition is an efficient temporal filtering method that explores the periodicity in time series, without the need to set filtering windows or assume profile shapes. It has been employed in reconstructing optical or radar observations to detect seasonal vegetation phenology [28] and flood events [29], [30]. The parameters derived from harmonic decomposition encompass information about both local and overall trends in the time series. Utilizing these parameters to recognize features would help avoid tedious local trend analysis. While existing research has integrated harmonic parameters into the statistical prediction model for crop classification [31], there remains a shortage of quantitative analysis on the correspondence between parameter compositions and seasonal patterns.

Rice-crop intensity varies in Southern China, with single or double rice-crop intensity being typical within the subtropical climate zone and triple rice-crop intensity occurring in the tropical zone [2], [32], [33]. However, it is challenging to summarize universal farming patterns for different rice-crop intensities because the quantity and timing of rice growth cycles in a field are influenced not only by natural conditions but also by the decisions made by individual farmers. In other words, rice phenology is asynchronous across different fields [16], [34]. The growth processes of rice and nonrice crops occurring at different times in the same field may be indistinguishable without the guidance of previous phenological information, as they sometimes exhibit similar backscatter variations, potentially leading to overestimations of rice-crop intensity. However, previous phenological information is unavailable when considering diverse local, regional, or national cultivation practices. With a limited understanding of the rice phenological diversity and radar backscatter complexity, a flexible approach for mapping rice-crop intensity nationwide is yet to come.

To address the three issues above in utilizing SAR data for large-scale rice-crop intensity mapping, we propose a novel scheme that comprises: 1) determining reliable features potentially related to rice growth under various cultivation practices; 2) employing efficient filtering and feature detection considering time-series periodicities; and 3) identifying rice growth cycles without the guidance of prior phenological information. Specifically, we develop a new harmonic-based method that effectively suppresses the time-series interference and recognizes reliable trough features within diverse rice backscatter patterns. Additionally, we propose a new temperature analysis method that consists of temperature restriction and overestimation correction to parse the potential rice phenology and distinguish nonrice backscatter features. The scheme is applied to map rice-crop intensity in Southern China in 2020. Sample sites located in different agro-climatic regions are selected to conduct field investigations, during which sample fields are collected for regional-level accuracy validation. Agricultural census data and the existing products related to rice-crop intensity of Southern China are used, respectively, for provincial-scale and national-scale comparison.

II. Study Area and Datasets

A. Study Area and Sample Sites

Fig. 1 shows Southern China, defined by South of the Qinling Mountains and Huaihe River line, North of the South China Sea, and West of the Pacific, extending westward to Sichuan and Yunnan provinces. The area is ∼2.6 million km^2 and is home to ∼680 million people, accounting for 25% of the national land area and 55% of the total population. Topographically, the area consists of mountains, rolling hills, and plains. The Qinling Mountains and Huaihe River line is a climate boundary dividing China into Northern and Southern China.

The climate in Southern China is dominated by tropical and subtropical monsoons, with high temperatures and heavy monsoon rains in the summer. The winter in most of Southern China is cold (with a low temperature near 0 °C) and dry, whereas areas near or below the Tropic of Cancer are warm (with a low temperature ≥10 °C) and rainy throughout the year. The temperature in Southern China usually reaches the maximum in July and the minimum in December.

A rice growth process can be roughly divided into seedling, transplanting, vegetative, reproductive, and mature periods.

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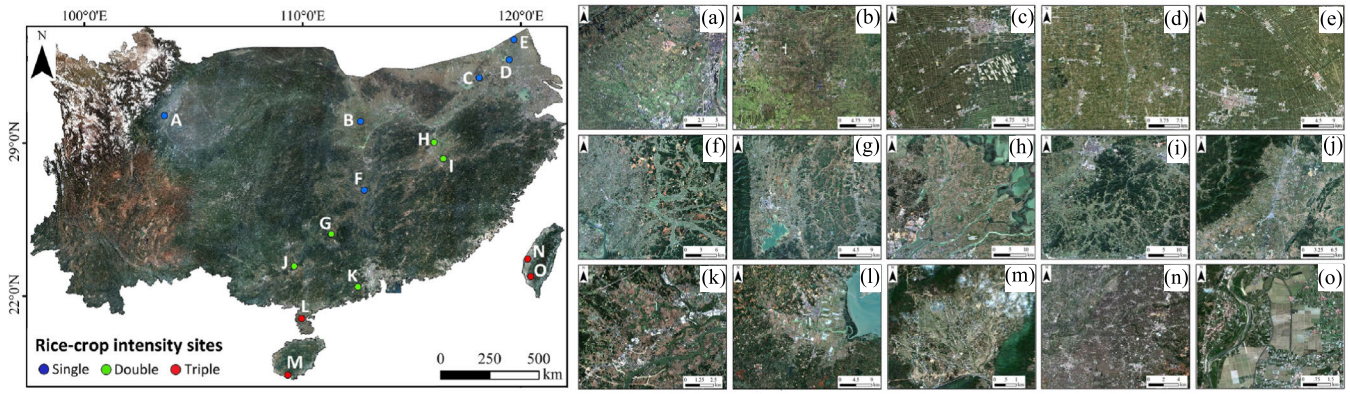


Fig. 1. Southern China defined in this study and 15 sample sites (A–O). The annual minimum value compositing of Sentinel-2 datasets is the background image.

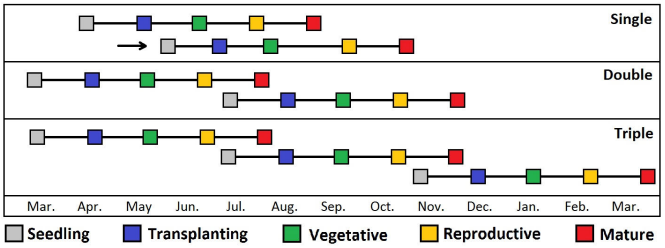


Fig. 2. Typical phenological calendar of the single, double, and triple rice-crop intensities in Southern China.

The process lasts 150–180 days, and each period lasts about 30 days in Southern China. The difference of 150–180 days usually occurs in the vegetative and reproductive periods, during which the development of rice vegetative organs (i.e., roots, stems, and leaves) and reproductive organs (i.e., flowers and ears) is influenced by various factors such as rice variety, climate conditions, and agricultural practices, leading to these two periods last a total of 60–90 days. When investigating remote sensing dynamics, the seedling period is not included in a rice growth cycle because satellites cannot monitor the period as the seedlings grow in another field.

Regional temperature is crucial for rice growth. Planting rice is usually prioritized over nonrice processes when the temperature is high. As reported by existing studies [35], [36], rice varieties in Southern China typically need ambient temperatures equal to or greater than 10 °C from the seedling period to the mature period for developing vegetative organs while accumulating photosynthetic products in the vegetative period or developing reproductive organs in the reproductive period requires an 18 °C temperature or higher.

According to our field investigations, Fig. 2 shows the typical rice phenology in one calendar year. The single rice-crop intensity traditionally dominates in the subtropical zone. The duration of warm ($\geq$10 °C, D_W) and hot ($\geq$18 °C, D_H) periods is usually insufficient for the double rice-crop intensity but longer than the days required for the single rice-crop intensity. Thus, the rice transplanting time can range from April (middle rice) to June (late rice), resulting in the harvest time from September to November. The arrow indicates the time variation. Since the highest temperatures in Southern China generally occur around July, middle rice usually experiences typical temperatures, which increase from seedling to harvest. In contrast, late rice experiences atypical temperatures, which decrease from transplanting to harvest.

The double rice-crop intensity is farther south, with longer D_W and D_H. The first crop is typically transplanted in the middle of March and harvested in late July. Then, the second crop is transplanted in early August and harvested in November. The triple rice-crop intensity is limited to tropical areas, where temperatures are $\geq$10 °C throughout the year and $\geq$18 °C from February to November. The transplanting times are usually April, August, and December. It should be cautioned that Fig. 2 should not be considered all-inclusive since rice is hardly farmed uniformly in Southern China. Farmers can more freely decide the rice transplanting time and the number of rice-crop cycles for the area with sufficiently long D_W and D_H.

Fifteen rice sample sites (A–O) of the primary grain production bases are selected. The single rice-crop intensity sites are located in the western (site A in Sichuan Basin), central (site B in Jianghan Plain, and site F in Hengyang Hills), and eastern (sites C, D, and E in Jianghuai Plain) zones. The double rice-crop intensity areas are located in the central (site H in Ganjiang Plain, and site I in Yingtan Hills) and southern (site G in Hezhou Hills, site J in Guigang Plain, and site K in Kaiping Plain) zones. The triple rice-crop intensity practices are located in the southernmost (site L in Leizhou Peninsula, site M in Tianya Basin, site N in Jianan Plain, and site O in Pingdong Plain) zones. Field investigation is conducted at sites A, B, C, G, H, and L to record the information about crop types, phenological periods, and field managements in different seasons. In addition, optical image interpretation is conducted using multitemporal Sentinel-2 and Gaofen-1/2/6 datasets to collect more field samples. About 300 sample fields are randomly selected per site, considering the areal proportion of nonrice fields, single, double, and triple rice-crop fields. Given that the size of a field in Southern China varies from 10 × 10 to 100 × 100 m, a sample field with the minimum size corresponds to one pixel, while a larger sample field corresponds to 10 or more pixels.

B. SAR and Auxiliary Datasets

Since the long time-series observation at high spatial resolution and a large scale are significantly computing resource-intensive, SAR and auxiliary data access and

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processing are performed on the Google Earth Engine (GEE) platform (https://developers.google.cn/earth-engine). The Sentinel-1 SAR data provided by GEE is analysis-ready data with a resolution of 10 m. Each scene has been preprocessed with Sentinel-1 Toolbox using the following steps: thermal noise removal, radiometric calibration, and terrain correction. The revisit period of Sentinel-1A/B is 12 days in Southern China, and it can be 6 when both Sentinel-1A&B are considered (https://dataspace.copernicus.eu/). In regions where swaths from neighboring orbits overlap, the revisit period can be further reduced. Multitemporal SAR ground range detected (GRD) data in 2020 are acquired, covering Southern China. The data acquisition mode is the interferometric wide swath (IW), with a swath of 250 km. All data are acquired along ascending orbits. Backscatter in logarithmic scale (dB) is converted into linear scale to ensure all mathematical operations on the SAR datasets maintain clear and desired physical significance. Then, time-series VH backscatter data are created.

The land use and land cover information of Southern China is derived from the European Space Agency (ESA) WorldCover product (https://esa-worldcover.org/en). The product provides a global land cover classification map in 2020 at 10-m resolution using a gradient-boosting decision tree algorithm based on Sentinel-1 and Sentinel-2 data. It contains 11 land cover classes, including cropland, but has no rice class. However, the cropland class delineates whether an area is a potential rice field.

The National Aeronautics and Space Administration (NASA) digital elevation model (DEM for short) is derived from the Land Processes Distributed Active Archive Center (LP DAAC, https://lpdaac.usgs.gov/products/nasadem_hgtv001), located in Sioux Falls, USA. The NASADEM has a cell size of 30 $\times$ 30 m. The altitude and slope derived from the NASADEM delineate the suitable geographical areas for rice cultivation.

The ambient air temperature data at 2 m above the ground surface are derived from the ERA5-Land product (https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5-land). The ERA5-Land combines the modeled data with multisource atmospheric observations. The temperature data are averaged daily with a spatial resolution of 9 $\times$ 9 km. A bilinear interpolation technique is used to resample the NASADEM and ERA5-Land datasets to a 10-m resolution.

Agricultural census data on rice crop intensity are derived from the Ministry of Agriculture and Rural Affairs of the People's Republic of China (http://www.moa.gov.cn/). The census data in 2020 are incomplete, and only the statistics for nine provinces (i.e., Henan, Hunan, Jiangsu, Jiangxi, Shanghai, Sichuan, Zhejiang, Fujian, and Hainan) are accessible for single or double rice-crop intensities. Data for triple rice-crop intensity are not officially recorded due to their minimal national contribution.

III. Method

A. Research Scheme

Fig. 3 shows the workflow of this study implemented on the GEE platform, consisting of the SAR/auxiliary data processing, harmonic analysis, rice growth cycle identification, and mapping/validation. Sentinel-1 datasets are collected and preprocessed to generate the VH backscatter time series. The short-term disturbances are suppressed while the essential trends are extracted based on harmonic decomposition. The fluctuation profiles and troughs of the backscatter variations are identified through a harmonic feature recognition method. Rice growth cycles manifested in time series are identified based on temperature restriction and overestimation correction. Then, the mapping of rice-crop intensity in Southern China is achieved through pixelwise counting rice growth cycles. This process is complemented by applying a terrain mask to retain low-altitude flat regions and a land cover mask to preserve cropland areas. Field samples are used for regional-scale accuracy validation and error analysis. The agricultural census data and existing products are used, respectively, for provincial-scale and national-scale comparison.

B. VH Backscatter Within a Rice Growth Cycle

The VH backscatter time series is analyzed in this study due to its sensitivity to rice biophysical changes and soil physicochemical variations. Rice is usually planted under typical temperatures, with a temperature-increasing trend from transplanting to harvest and exceeding 25 °C during the reproductive period. As shown in Fig. 4(a), rice backscatter reaches a trough during the transplanting period due to the sparse plant coverage and smooth water surface after flooding. The backscatter crests during the vegetative or reproductive period due to the dense rice canopy. The backscatter pattern would repeat two or three times for an area with a double or triple rice-crop intensity.

A noticeable exception exists, as shown in Fig. 4(b), when farmers plant late rice near the northeastern part of Southern China under atypical temperatures, with temperatures exhibiting a declining trend from transplanting to harvest and approaching 18 °C during the reproductive period. The transplanting period starts in late June when the temperature is usually hot. Rice canopies develop rapidly, leading to the backscatter increases quickly and substantially from transplanting to vegetative periods. As a result, the backscatter trough during the transplanting period is a short-term fluctuation. Due to the late start, the reproductive period will likely encounter a colder autumn. Keeping rice fields flooded is standard to maintain the temperature. As a result, strong attenuation caused by the smooth water surface occurs in the reproductive period, bringing a most remarkable and extended trough. Despite the difference under different temperature types, the backscatter presents quasi-cosine variations, and one significant backscatter trough exists within one rice growth cycle. Thus, a time-series modeling method that considers this mechanism is required to filter out short-term negligible troughs and preserve long-term substantial troughs. Then, identifying the rice-crop intensity can be transformed into detecting the troughs.

C. Harmonic Decomposition

Time-series modeling aims to construct interpretive models of annual SAR observations without overfitting, flexibly

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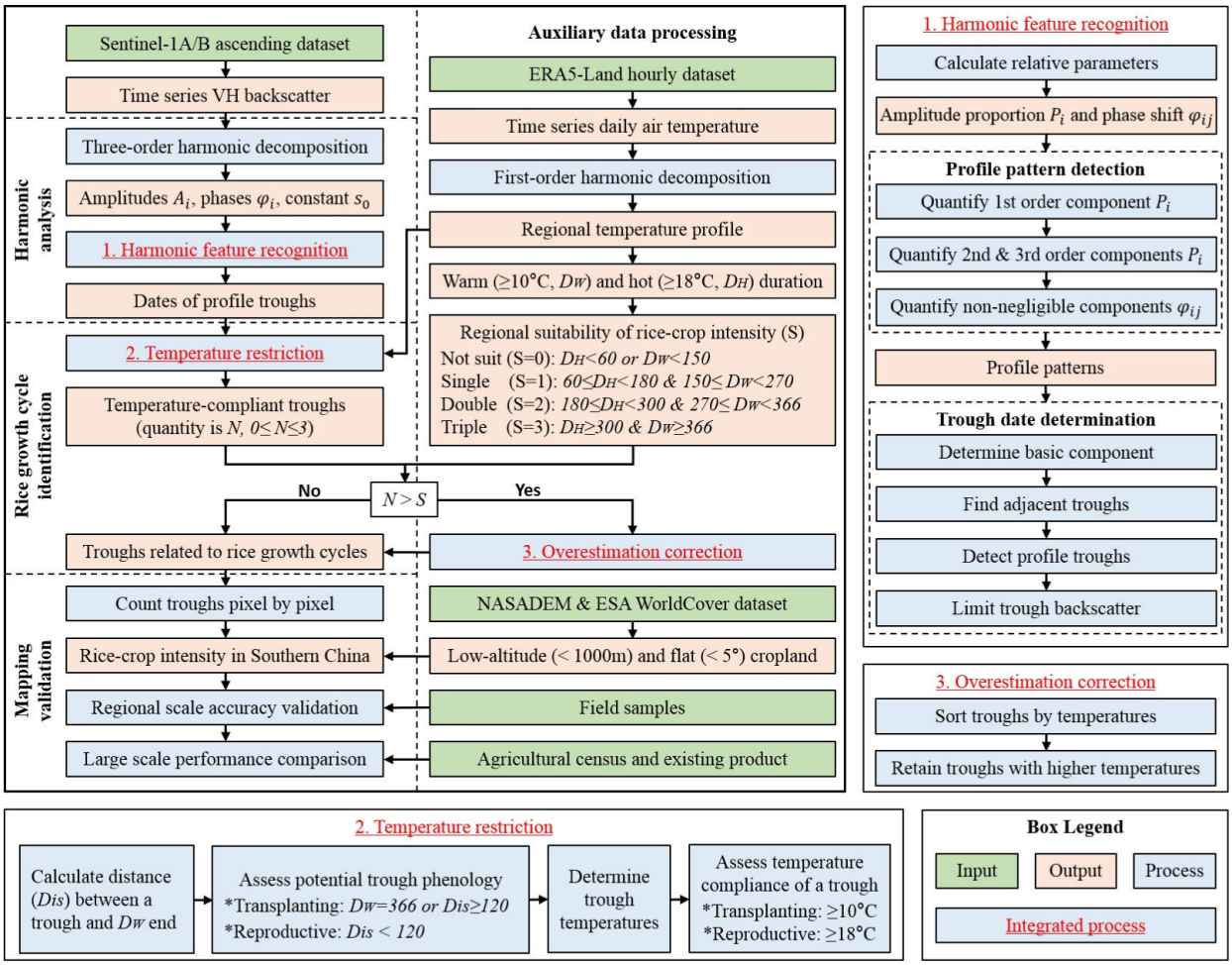
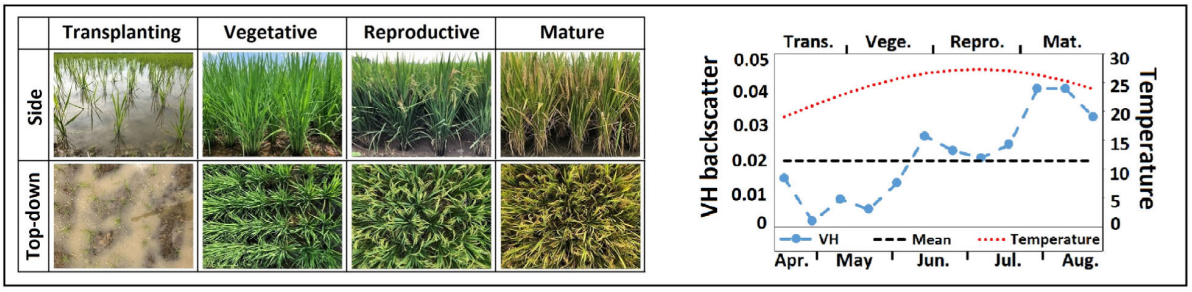
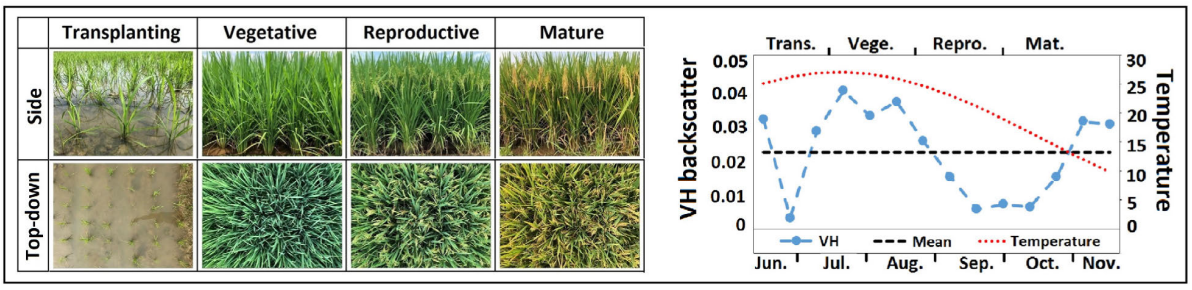


Fig. 3. Flowchart and its three key technique modules for this study.



(a)



(b)

Fig. 4. Rice phenology evolution and corresponding backscatter and temperature variations within one rice-growth cycle under (a) typical and (b) atypical temperatures.

utilizing simple and replicable shapes to explain seasonal trends. Here, the harmonic decomposition method explores the periodic characteristics based on the Fourier analysis [31], [37], decomposing time-series data into a constant component and a set of cosine components. The quasi-cosine backscatter variations repeat once, twice, and thrice in the fields with

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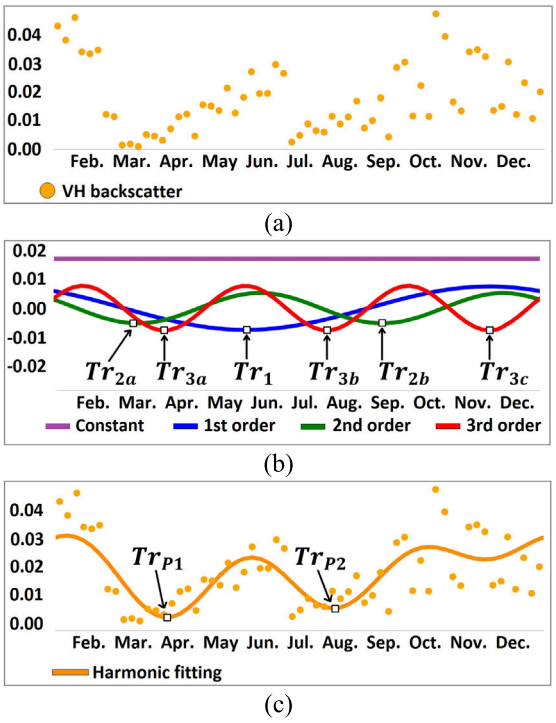


Fig. 5. Example of harmonic decomposition for the annual backscatter from a field with double rice-crop intensity. (a) Original time-series backscatter. (b) Harmonic components. (c) Harmonic fitting. Troughs on the curves are labeled.

single, double, and triple rice-crop intensities. Thus, cosine components up to three orders essentially describe the annual backscatter fluctuation. The components describe the short- and long-term periodical characteristics of land processes, providing the basis to interpret the time series. Their synthesis profile (i.e., harmonic fitting) is a smooth curve that reconstructs the original data, as shown in Fig. 5.

It should be noted that the Fourier analysis mentioned involves decomposing data in the time domain using the Fourier series rather than employing the more intricate frequency-domain Fourier transform. The annual SAR observations are treated as a single cycle of a periodic signal, fulfilling the requirement that the Fourier series exclusively pertains to periodic signals [29], [30]. The mathematical implementation of harmonic decomposition is introduced as follows.

Let $S[t]$ represent the time-series VH backscatter data of one year, where t is the time measured as the day of the year (DOY). To simplify the analysis, we normalize t into the 0–1 range by the total number of days (i.e., 366) in 2020. With the harmonic decomposition in this study, one writes

$$\begin{aligned} S[t] &= s_0 + \sum_{i=1}^{3} s_i[t] + s_r[t] \\ &= s_0 + \sum_{i=1}^{3} A_i \cos\left[2\pi\omega_i t - \varphi_i\right] + s_r[t] \end{aligned} \tag{1}$$

where s_0 is the zeroth order constant component, quantifying the mean within the $S[t]$. The ith-order component $s_i[t]$ corresponds to the cosine variations repeated ω_i ($\omega_i = i$) times annually. $s_r[t]$ is a residual component, including all other higher order ($i \geq 4$) components, and is treated as noise. The ω_i, A_i, and φ_i denote, respectively, the frequency, amplitude, and phase of $s_i[t]$. φ_i ranges from $-\pi$ to π. Following cosine expansion principles, $S[t]$ is further decomposed into a constant term and a series of cosine and sine functions

$$S[t] = s_0 + \sum_{i=1}^{3} \left(b_i \cos\left[2\pi\omega_i t\right] + c_i \sin[2\pi\omega_i t]\right) + s_r[t] \tag{2}$$

where $b_i = A_i \cos\varphi_i$ and $c_i = A_i \sin\varphi_i$. The expression corresponds to the typical form of a Fourier series. The s_0, b_i, and c_i are unknown and are determined with the linear least-squares regression, where the constant, cosine, and sine terms are taken as the basis functions for fitting the $S[t]$. Then, A_i and φ_i are obtained as

$$A_i = \sqrt{\left(b_i^2 + c_i^2\right)}, \quad \varphi_i = \arctan\left(c_i / b_i\right). \tag{3}$$

At this point, $S[t]$ can be transformed into the synthesis of s_0 and three order $s_i[t]$ with determined ω_i, A_i, and φ_i. Troughs in the first order (e.g., Tr_1 in Fig. 5), second-order (e.g., $\mathrm{Tr}_{2a/b}$), and third order (e.g., $\mathrm{Tr}_{3a/b/c}$) $s_i[t]$ are indicative of periodic declines of the backscatter variations, and the trough dates can be calculated as follows:

$$\begin{cases} \mathrm{Tr}_1 = 366(\pi + \varphi_1)/2\pi \\ \mathrm{Tr}_{2a} = 366(\pi + \varphi_2)/2\pi - 366/4 \\ \mathrm{Tr}_{2b} = 366(\pi + \varphi_2)/2\pi + 366/4 \\ \mathrm{Tr}_{3a} = 366(\pi + \varphi_3)/2\pi - 366/3 \\ \mathrm{Tr}_{3b} = 366(\pi + \varphi_3)/2\pi \\ \mathrm{Tr}_{3c} = 366(\pi + \varphi_3)/2\pi + 366/3. \end{cases} \tag{4}$$

The residual component $s_r[t]$ quantifies the differences between the VH backscatter and the fit value. To evaluate the fitting performance, the root-mean-square error (RMSE_r) of harmonic decomposition can be calculated based on $s_r[t]$

$$\mathrm{RMSE}_r = \sqrt{\frac{\sum_{t=1}^{N} (s_r[t])^2}{N}} \tag{5}$$

where N is the number of observations (data points in the time series).

D. Harmonic Feature Recognition

The prerequisite for identifying the rice-crop intensity is to determine the number and timing of profile troughs (e.g., Tr_{p1} and Tr_{p2} in Fig. 5). The determination can be transformed into determining the profile pattern (single-, double-, or triple-trough pattern) and its position along the time axis. A harmonic feature recognition method is proposed to efficiently detect profile troughs based on the cosine parameters without performing time-series traversal operations. The method consists of profile pattern detection and trough date determination, as shown in Fig. 3. Since cosine components are periodic functions, the profile pattern of their synthesis can be determined based on their relative parameters

$$\begin{cases} P_i = A_i / \sum_{n=1}^{3} A_n \\ \varphi_{ij} = \varphi_j - j\varphi_i / i \end{cases} \tag{6}$$

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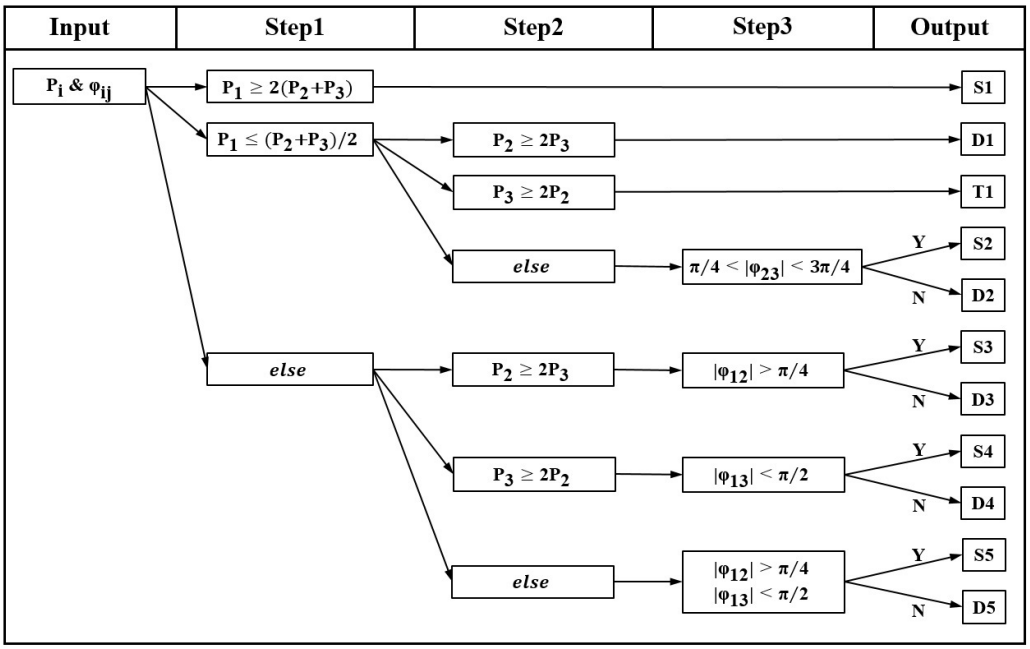


Fig. 6. Detecting profile patterns of annual backscatter variations.

where P_i quantifies the amplitude proportion of $s_i[t]$, and φ_{ij} describes the phase shift of $s_j[t]$ relative to $s_i[t]$. φ_{ij} ranges from $-\pi$ to π. Synthesis analysis of three components with diverse P_i and φ_{ij} is quite challenging because their interactions are intricately complex. However, the analysis can be simplified if one or two components are negligible due to low amplitude proportion. In signal processing, it is a common practice to use the 3-dB cutoff points to define the effective frequency range in a signal's spectrum [38], implying that components can be considered negligible when their amplitudes fall below half of the amplitude of the component of interest.

A rice growth cycle (from transplanting to mature period) typically lasts for 120–150 days, falling between the periods of $s_2[t]$ (183 days) and $s_3[t]$ (122 days), and is significantly shorter than that of $s_1[t]$ (366 days). Therefore, the periodicity of rice backscatter variations is primarily described by $s_2[t]$ and $s_3[t]$, while $s_1[t]$ mainly describes the annual trend, unless it has a dominant amplitude. Hence, synthesis analysis (i.e., profile pattern detection) is carried out in three steps, as shown in Fig. 6.

1) *Quantifying $s_1[t]$ Contribution:* If $s_1[t]$ amplitude is less than half of the higher order ($s_2[t]$ and $s_3[t]$) amplitude ($P_1 \leq (P_2 + P_3)/2$), $s_1[t]$ is negligible while the entirety of $s_2[t]$ and $s_3[t]$ has a dominant contribution. Conversely, if the higher order amplitude is less than half of $s_1[t]$ amplitude [$P_1 \geq 2(P_2 + P_3)$], $s_1[t]$ has a dominant contribution while the entirety of $s_2[t]$ and $s_3[t]$ is negligible. In other cases, $s_1[t]$ is nonnegligible, and the entirety of $s_2[t]$ and $s_3[t]$ is also nonnegligible.
2) *Quantifying $s_2[t]$ and $s_3[t]$ Contribution:* This step is performed when the entirety of $s_2[t]$ and $s_3[t]$ is nonnegligible or has a dominant contribution. If $s_3[t]$ amplitude is less than half of $s_2[t]$ amplitude ($P_2 \geq 2P_3$), $s_2[t]$ is nonnegligible. Conversely, if $s_2[t]$ amplitude is less than half of $s_3[t]$ amplitude ($P_3 \geq 2P_2$), $s_3[t]$ is nonnegligible. In other cases, both $s_2[t]$ and $s_3[t]$ are nonnegligible.
3) *Quantifying Phase Shift of Nonnegligible Components:* If only one component is nonnegligible (patterns S1, D1, and T1), the profile pattern is dominated by that component, and phase-shift analysis is not needed. If only $s_2[t]$ and $s_3[t]$ are nonnegligible, their synthesis profile belongs to a single-trough pattern when a trough in $s_2[t]$ is synchronized ($\pi/4 < |\varphi_{23}| < 3\pi/4$) with a trough in $s_3[t]$ (pattern S2). Otherwise, the profile belongs to a double-trough pattern (pattern D2). If only $s_1[t]$ and $s_2[t]$ are nonnegligible, their synthesis profile belongs to a single-trough pattern when the trough in $s_1[t]$ is synchronized ($|\varphi_{12}| > \pi/4$) with a trough in $s_2[t]$ (pattern S3). Otherwise, the profile belongs to a double-trough pattern (pattern D3). If only $s_1[t]$ and $s_3[t]$ are nonnegligible, their synthesis profile belongs to a single-trough pattern when the trough in $s_1[t]$ is synchronized ($|\varphi_{13}| < \pi/2$) with a trough in $s_3[t]$ (pattern S4). Otherwise, the profile belongs to a double-trough pattern (pattern D4). If all the components are nonnegligible, their synthesis profile belongs to a single-trough pattern when the trough in $s_1[t]$ is synchronized ($|\varphi_{12}| > \pi/4$, and $|\varphi_{13}| < \pi/2$) with both a trough in $s_2[t]$ and a trough in $s_3[t]$ (pattern S5). Otherwise, the profile belongs to a double-trough pattern (pattern D5).

In summary, the profile patterns are determined by nonnegligible components and their relative positions. Patterns S1–S5 represent one fluctuation annually. For pattern S1, the profile trough is characterized by the trough in $s_1[t]$. For patterns S2–S5, troughs in two or three nonnegligible components overlap, jointly forming a profile trough. Patterns D1–D5 represent two fluctuations annually. For patterns D1 and D3, the two profile troughs are characterized by the two troughs in $s_2[t]$. For pattern D4, the two profile troughs are characterized

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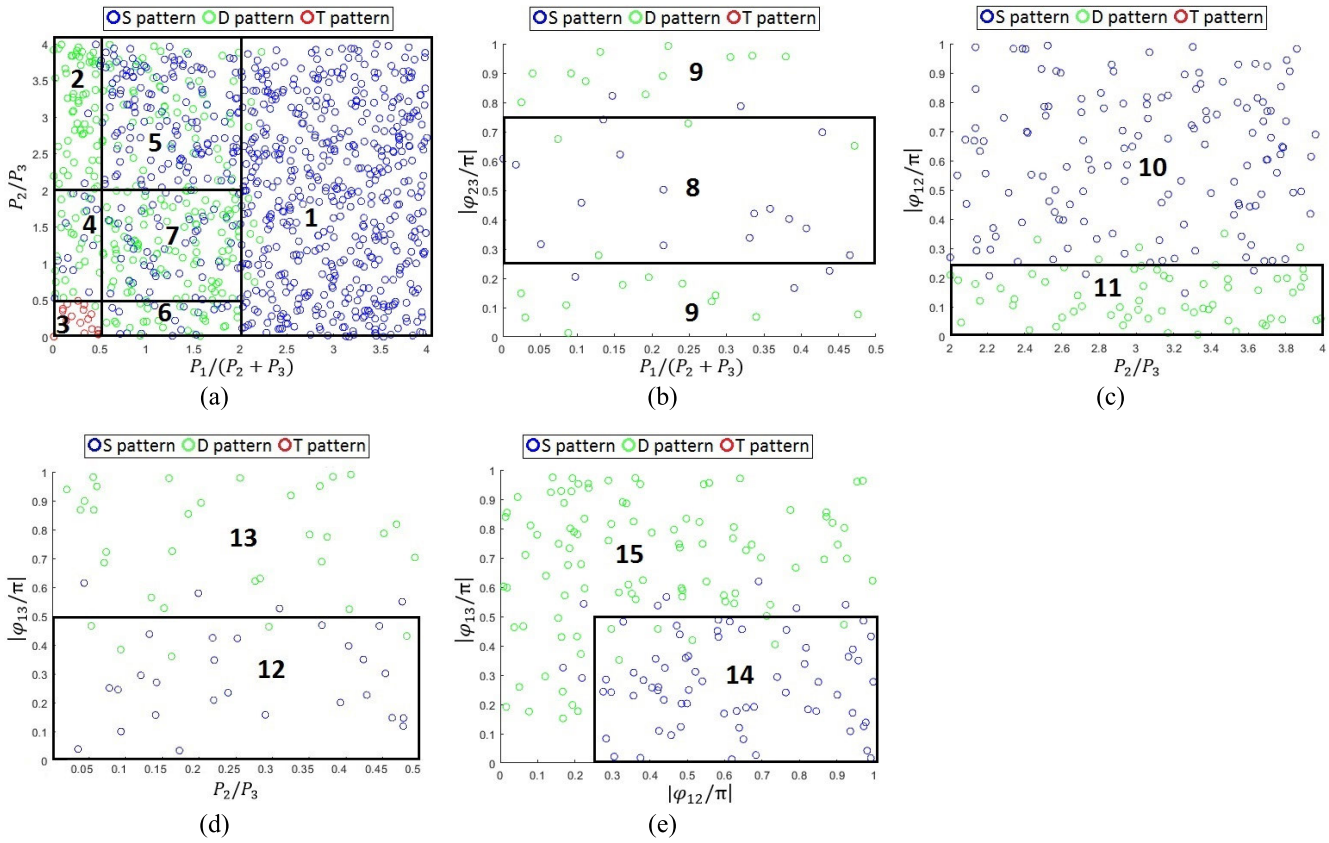


Fig. 7. Randomly generated data points illustrating pattern separability and segmentation thresholds in five feature planes, where zones 1–3 and 8–15 correspond, respectively, to patterns: (a) S1, D1, and T1; (b) S2 and D2; (c) S3 and D3; (d) S4 and D4; and (e) S5 and D5.

by two troughs in $s_3[t]$. For patterns D2 and D5, the two profile troughs are characterized jointly by the two troughs in $s_2[t]$ and two troughs in $s_3[t]$. In patterns D3–D5, the trough in $s_1[t]$ does not overlap with any troughs in $s_2[t]$ and $s_3[t]$. Therefore, $s_1[t]$ does not directly shape the profile troughs but describes the overall trend of the time series. Pattern T1 represents three fluctuations annually, and the three profile troughs are characterized by the three troughs in $s_3[t]$. The amplitudes of nonnegligible components are generally similar. When the amplitudes change, the range of φ_{ij} for characterizing the profile pattern may exhibit slight variations. Nevertheless, for the sake of method simplification, this study has chosen to disregard these minor differences. Please refer to the Appendix for the details of phase-shift analysis and spatial distributions of harmonic parameters.

Separability analysis of profile patterns within the feature space can further confirm the rationality of the constructed decision tree, as shown in Fig. 7. A total of 1000 sample sets of three-order cosine components with different amplitudes and phases are randomly generated. The synthesis profiles of each set of three-order components are then identified as single (S), double (D), or triple (T) trough patterns based on the number of troughs with comparable depth. The depth is the difference between the trough's minimum value and the time-series' mean value. Two troughs have comparable depths when the depth of the shallower trough is one-half or more of the deeper trough.

Since the 6-D feature space is challenging to visualize, it is decomposed into five feature planes to illustrate the pattern separability and segmentation thresholds step by step. Each data point in a feature plane corresponds to a set of three-order cosine components. Points are assigned blue, green, or red when their corresponding profiles belong to S, D, or T patterns, respectively. Feature planes are partitioned into multiple zones with black borders, each labeled with numbers. A zone predominantly comprising points of the same color corresponds to a specific profile pattern.

In Fig. 7(a), points within zones 1–3 correspond to patterns S1, D1, and T1, respectively. Zones 4–7 exhibit a mix of blue and green points, requiring further separation in other feature planes. Points from zone 4 are then illustrated in Fig. 7(b), where zones 8 and 9 correspond to patterns S2 and D2, respectively. Similarly, points from zone 5 are depicted in Fig. 7(c), with zones 10 and 11 corresponding to patterns S3 and D3, respectively. Points from zones 6 and 7 are represented in Fig. 7(d) and (e), with zones 12–15 corresponding to patterns S4, D4, S5, and D5, respectively.

To get the timing of troughs for each profile pattern, we propose a trough date determination method to eliminate the need for laborious inflection point detection. The method includes four steps.

1) *Determining the Basic Component:* The component should indicate the number and approximate time of occurrence for profile troughs. The component is $s_1[t]$ for patterns S1–S5, $s_2[t]$ for patterns D1–D5, and $s_3[t]$ for pattern T1.
2) *Finding Adjacent Troughs:* For each trough in the basic component (e.g., Tr_1 for pattern S5 in Fig. 8), two

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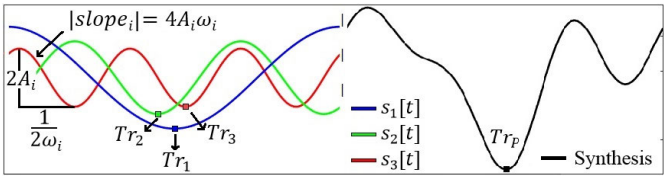


Fig. 8. Profile trough and corresponding three troughs in cosine components.

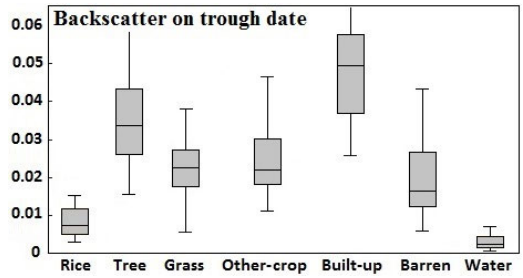


Fig. 9. Backscatter on trough date for different land cover types.

adjacent troughs in the other two cosine components are found (e.g., Tr_2 and Tr_3 in Fig. 8).

3) *Detecting Profile Troughs:* A profile trough is the decrease over a specific duration. The decrease rate of the profile depends on that of the three $s_i[t]$. The decrease rate of $s_i[t]$ can be approximated by the absolute value of the crest-trough slope (i.e., $4A_i\omega_i$ in Fig. 8). The higher the amplitude (A_i) or frequency (ω_i) of $s_i[t]$, the steeper its slope, resulting in a more substantial contribution to a profile trough. Thus, the date of a profile trough is determined by calculating the weighted average of the troughs in the three $s_i[t]$, whose weight is $A_i\omega_i$

$$\mathrm{Tr}_P = \sum_{i=1}^{3} A_i\omega_i \mathrm{Tr}_i \Big/ \sum_{i=1}^{3} A_i\omega_i. \tag{7}$$

4) *Limiting Trough Backscatter*: According to our sample statistics (Fig. 9), rice backscatter troughs, which may related to rice transplanting or reproductive period, are generally close to 0.01 (−20 dB) due to the smooth underlying surface or extremely dense canopy. In comparison, those in other dry/rough cropland are usually higher than 0.02 (−17 dB). Thus, a lenient threshold of 0.02 is determined to eliminate troughs that are evidently irrelevant to rice growth.

Statistical samples are derived from 3990 sample fields located in 15 sample sites. Among these, there are 1503, 942, and 300 sample fields with single, double, and triple rice-crop intensities, respectively, and 1245 samples of nonrice cropland. To supplement the dataset, 2574 samples of noncropland derived from ESA WorldCover product are included. This supplementary set comprises samples from other land cover types, including trees (526), grass (453), built-up (696), barren (387), and water (512) areas. Given the presence of multiple troughs in the annual backscatter time series, particularly in samples with multiple rice-crop intensities, a total of 12 170 trough samples are obtained for statistical analysis. This includes 3266 troughs related to the rice transplanting period, 1021 troughs related to the rice reproductive period, 2257 troughs related to nonrice crops, and 5626 troughs related to land processes in noncropland areas.

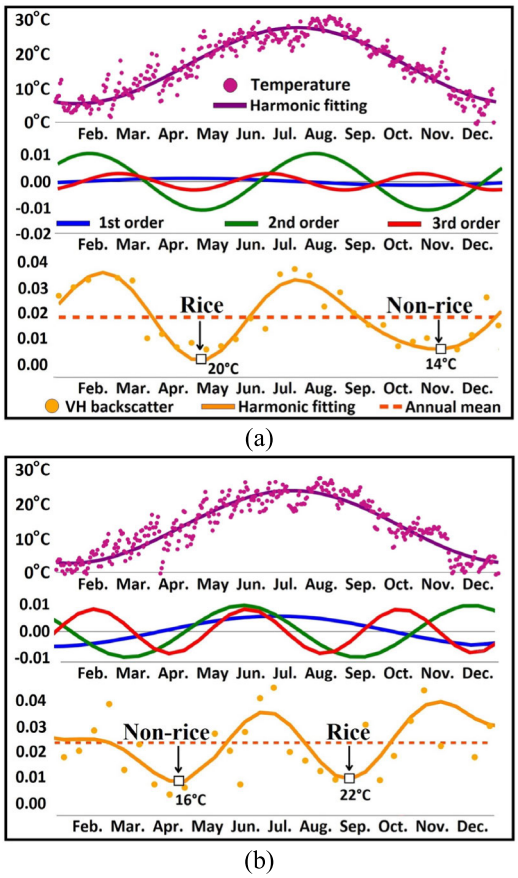


Fig. 10. Temperature and backscatter time series derived from (a) site B and (b) site C. In each plot, the first row is temperature variations and their harmonic fitting, and the second and third rows are cosine components and harmonic fitting of the VH backscatter variations, respectively.

E. Rice Growth Cycle Identification

Although deep troughs occurring in rice fields are usually related to rice cultivation, some may relate to unexpected backscatter reduction caused by rainfall waterlogging, artificial irrigation, land leveling, or canopy attenuation of nonrice crops beyond the rice growth cycles. As illustrated in Fig. 10, fields with double trough patterns are identified at site B [Fig. 10(a)] and site C [Fig. 10(b)]. However, the troughs near December in Fig. 10(a) correspond to waterlogged fallows in winter, while the trough near April in Fig. 10(b) corresponds to the reproductive growth of spring wheat, as noted in the fieldwork. Without the guidance of prior phenological information, these nonrice troughs often cause the overestimation of rice-crop intensity. Here, temperature analysis is conducted to distinguish these nonrice troughs from the rice ones, including a temperature restriction procedure and an overestimation correction procedure.

Random noise may occur in the temperature time series due to short-term meteorological changes or climate model errors. Thus, the harmonic decomposition is also applied to reconstruct the daily temperature in Southern China. Considering the quasi-cosine feature of the temperature dynamics, we use

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a constant component and the first-order cosine component to fit the ERA5-Land temperature data in 2020. With the reconstructed temperature profiles, the duration of the warm period ($\geq$10 °C, D_W) and hot period ($\geq$18 °C, D_H) is determined, and the temperatures on backscatter trough dates are estimated.

Temperature restriction emphasizes that a backscatter trough is potentially related to rice growth when the temperature requirement of its corresponding rice phenological period is met. Specifically, if the temperatures in an area are $\geq$10 °C throughout the year ($D_W = 366$) or if the time interval (Dis) between the profile trough date and the end date of the warm period (e.g., near mid-November at site B and late-October at site C) exceeds 120 days (the shortest duration from transplanting to mature period), rice has sufficient time to complete one growth cycle. In this case, the trough potentially relates to the rice transplanting period when the temperature on the trough date is higher than 10 °C (e.g., troughs before July in Fig. 10). Conversely, if Dis $<$ 120, rice may be planted under atypical temperatures. In this case, the trough is potentially related to the rice reproductive period when the temperature on the trough date is higher than 18 °C [e.g., the trough after July in Fig. 10(b)]. Troughs meeting the temperature restriction are defined as temperature-complaint troughs.

Temperature-complaint troughs may still correspond to nonrice land processes [e.g., the trough before July in Fig. 10(b)] when D_W and D_H in an area is shorter than what is required for a high rice-crop intensity, but longer than what is required for a lower rice-crop intensity. In this case, farmers often plant nonrice crops (e.g., wheat or canola) before or after rice cropping. Planting nonrice crops can sometimes lead to deep backscatter troughs, and the temperature on nonrice trough dates may also meet rice growth requirements. Thus, these nonrice troughs would be misidentified as rice ones, causing the overestimation of rice-crop intensity. To correct the overestimation, we compare the trough temperatures. A rice growth cycle typically experiences more favorable temperature conditions than nonrice crops. Thus, once the maximum suitable rice-crop intensity for an area is known, the number of troughs exceeding this maximum is identified. These extra troughs, characterized by unfavorable temperature conditions, are considered nonrice troughs and are subsequently removed. Then, only the rice troughs are retained, and the rice-crop intensity is identified by counting the number of rice troughs.

To determine the maximum suitable rice-crop intensity for different areas, we propose a geographical zoning guideline based on D_W and D_H. Considering the shortest duration (150 days) of a rice growth process from seedling to mature periods, with each period lasting 30 days, the single rice-crop intensity needs at least 60 days of D_H for the vegetative and reproductive periods, and at least 150 days of D_W throughout the process. For multiple rice-crop intensities, the former cycle's mature period usually coincides with the latter cycle's seedling period. The temperatures are $\geq$18 °C from the first cycle's vegetative period to the last cycle's reproductive period and are $\geq$10 °C from the first cycle's seedling period to the last cycle's mature period. Therefore, more than 180 days of D_H and 270 days of D_W are essential for a double rice-crop intensity. The triple rice-crop intensity needs 300-day D_H and 366-day D_W. Thus, a temperature suitability mask, including cold (unsuitable), mild (single rice-crop intensity), warm (double rice-crop intensity), and hot (triple rice-crop intensity) zones, is created.

F. Map Rice-Crop Intensity and Assess Accuracy

A topography mask based on the NASADEM removes high-altitude or nonflat areas. The high-altitude areas are not ideal for rice growth due to several factors, including low temperatures, large diurnal temperature variations, thin air, and suboptimal soil conditions characterized by low fertility and moisture [39]. Although a limited number of rice fields exist at altitudes $\geq$1000 m, they are not considered in this study. In addition, rice is mostly planted in flat areas because rice cropping demands a continuous water supply [40]. Although some terraced rice fields exist at slope $\geq$5° according to our investigation, they are also not considered in this study.

In addition, a cropland mask based on the ESA WorldCover product is created to remove the noncropland areas. Apart from the crop growth, some events may also cause significant backscatter fluctuations, including the phenological evolution of vegetation on shrubland, grassland, forest, and wetland, as well as the rainfall waterlogging on bare or sparse vegetation cover land, the melting of ice and snow, the flooding of land adjacent to water bodies during the rainy season, and the backscatter instability resulting from complex geometric structures of buildings. These events may cause backscatter troughs under high temperatures, impeding the accurate identification of the rice planting areas. Therefore, noncropland areas are removed using the cropland mask. According to the product validation report (https://esa-worldcover.org/en/data-access), the overall accuracy (OA) of the ESA WorldCover product achieved for 11 land cover classes in Asia is 80.7%. Specifically, the producer's accuracy (PA) and the user's accuracy (UA) for cropland are 82.1% and 80.5%, respectively.

In summary, harmonic analysis and rice growth cycle identification are implemented location by location (or cell by cell) within the low-altitude flat cropland. We apply a spatial mode filter with a 3 $\times$ 3 window to further refine the results. Then, the rice-crop intensity of Southern China is mapped.

To validate the accuracy of the rice-crop intensity product, we use sample fields in 15 sample sites. A field is deemed correctly identified when 70% of pixels within the field are correctly identified. Then, the user's, producer's, and overall accuracies quantify the accuracy levels. Agricultural census data are used for provincial-scale validation. In addition, the spatial consistency between our results and existing products in 2020 is evaluated at a national scale. The products include the 10 $\times$ 10 m single [41] and double [42] rice-crop intensity data.

IV. Results

A. Harmonic Features and Rice Growth Cycles

Fig. 11 shows the representative time-series VH backscatter for different rice-crop intensities. Fig. 11(a)–(c) is derived

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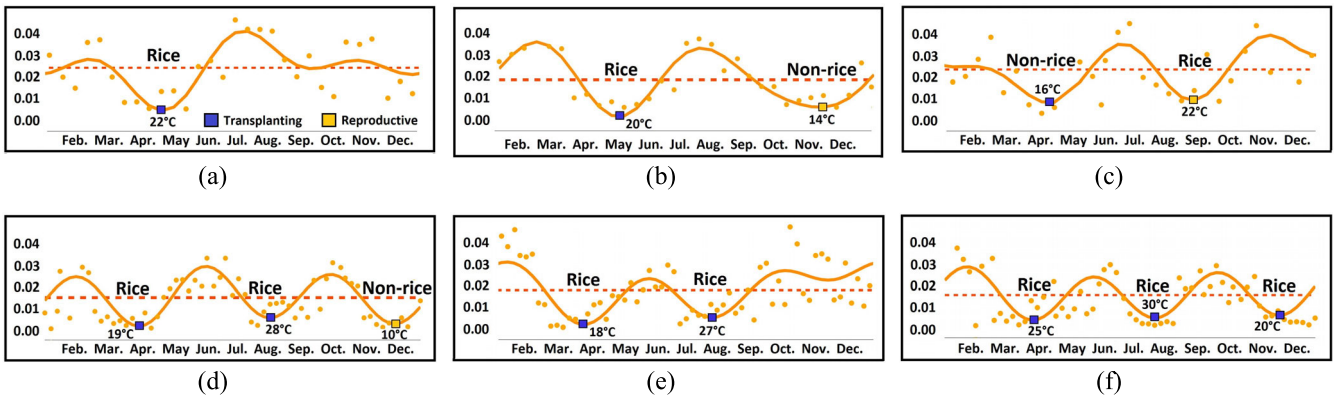


Fig. 11. Profile patterns and troughs derived from sites (a) A, (b) B, (c) C, (d) H, (e) G, and (f) L.

from rice fields in sites A–C, respectively. These sites are suitable for a single rice-crop intensity. Fig. 11(d) and (e) is derived from sites H and G, respectively, suitable for a double rice-crop intensity. Fig. 11(f) is derived from site L, where a triple rice-crop intensity is suitable. In ideal cases, rice fields with a single, a double, or a triple rice-crop intensity exhibit one [Fig. 11(a)], two [Fig. 11(e)], and three [Fig. 11(f)] deep troughs, respectively. Fig. 11(a), (e), and (f) corresponds to patterns S5, D5, and T1, respectively. In nonideal cases, nonrice land processes cause significant backscatter declines, leading to the presence of extra troughs in Fig. 11(b)–(d). As a result, the contribution of the cosine components changes. Fig. 11(b), (c), and (e) corresponds to patterns D1, D2, and T1. Differentiating nonrice troughs from rice ones based on backscatter characteristics is challenging because the variations are similar [see Fig. 11(d) and (f)].

Phenological attributes on trough dates are identified based on temperature restriction. The troughs near December in Fig. 11(b) and (d) and the trough near September in Fig. 11(c) potentially relate to the reproductive growth of the late rice, while the other troughs in Fig. 11 may relate to the rice transplanting period. In Fig. 11(a), (e), and (f), temperature requirements ($\geq$18 °C in the reproductive period and $\geq$10 °C in the transplanting period) are met for all troughs, and the number of troughs does not exceed the site's multiple rice-crop suitability. Thus, these troughs are identified as rice troughs. On the other hand, the troughs near December in Fig. 11(b) and (d) are identified as nonrice troughs related to winter waterlogged fallows. After removing these troughs, the number of the remaining ones in Fig. 11(b) and (d) aligns with the regional multiple rice-crop suitability. Therefore, the trough near May in Fig. 11(b) and the troughs near April and August in Fig. 11(d) are identified as rice troughs. As for Fig. 11(c), temperature requirements are met for both troughs, but only a single rice-crop intensity is suitable for site C. Therefore, overestimation is considered to occur. According to the overestimation correction, the trough near April is identified as a nonrice one due to its suboptimal temperature supply.

B. Rice-Crop Intensity in Southern China

According to the geographical zoning guideline, coupled with the masks of slope <5°, altitude <1000 m, and cropland area, a suitability mask of multiple rice-crop intensities is produced in Fig. 12(a), where the blue (mild), green (warm), and red (hot) areas indicate the suitability for single, double, and triple rice-crop intensities, respectively. The gray area is unsuitable for rice cultivation.

Harmonic analysis and rice growth cycle identification are conducted pixel by pixel with the suitability mask. Then, rice-crop intensity throughout Southern China is mapped. The product and area statistics are shown in Fig. 12(b). Fields with a single rice-crop intensity are primarily concentrated in the central and northern parts, constituting the dominant (87.54%) rice cultivation practice. Fields with a triple rice-crop intensity account for an extremely small proportion (0.17%) and are distributed in the southernmost hot zone, while fields with a double rice-crop intensity account for 12.29% of the total.

TABLE I
ACCURACY ASSESSMENT FOR EACH SITE

Site	UA-S	UA-D	UA-T	UA-N	PA-S	PA-D	PA-T	PA-N	OA
A	85.37%			78.79%	83.33%			81.25%	82.43%
B	88.10%			78.57%	86.05%			81.48%	84.29%
C	91.30%			76.00%	87.50%			82.61%	85.92%
D	91.84%			74.19%	84.91%			85.19%	85.00%
E	89.80%			77.78%	88.00%			80.77%	85.53%
F	88.89%			77.78%	80.00%			87.50%	83.33%
G	82.86%	77.78%		79.41%	78.38%	75.00%		87.10%	80.21%
H	81.82%	78.79%		83.33%	79.41%	81.25%		83.33%	81.25%
I	76.47%	82.86%		91.18%	83.87%	85.29%		81.58%	83.50%
J	80.00%	86.36%		81.48%	80.00%	86.36%		81.48%	83.02%
K	78.57%	90.00%		80.77%	84.62%	81.82%		87.50%	84.29%
L	76.47%	79.49%	76.00%	84.00%	76.47%	75.61%	82.61%	84.00%	79.25%
M	76.47%	82.76%	76.47%	83.33%	72.22%	75.00%	83.87%	85.71%	80.17%
N	78.95%	85.37%	78.57%	70.00%	75.00%	85.37%	68.75%	82.35%	79.79%
O	84.21%	82.35%	75.76%	83.33%	80.00%	84.85%	83.33%	71.43%	80.77%

C. Accuracy Assessment

Fig. 13 shows the single, double, and triple rice-crop intensities in blue, green, and red at 15 sample sites. Only the single rice-crop intensity occurs at sites A–F. The single and double rice-crop intensities are mixed at sites G–K. The single, double, and triple rice-crop intensities are mixed at sites L–O.

Then, field-collected samples of multiple rice-crop intensities and nonrice cropland are used for accuracy assessment, quantifying UA, PA, and OA for single (S), double (D), triple (T) rice-crop intensities and nonrice (N) cropland, as shown in Table I.

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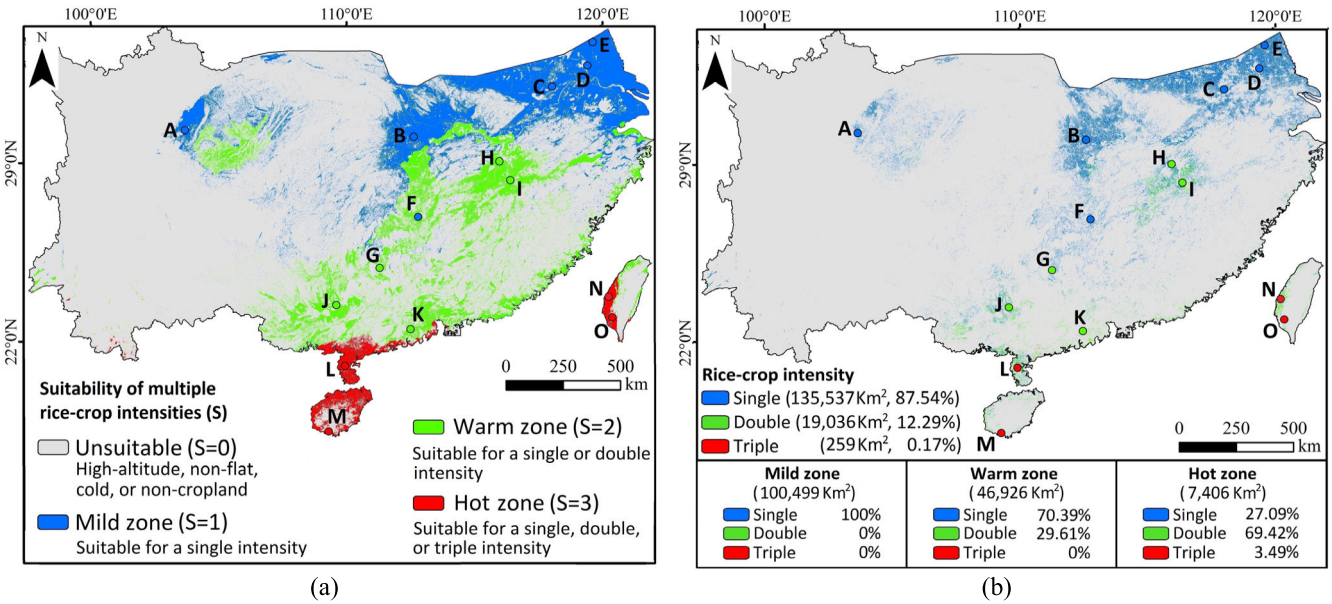


Fig. 12. (a) Suitability mask of multiple rice-crop intensities and (b) rice-crop intensity in Southern China, where the proportions of the single, double, and triple rice-crop intensities within different suitability zones are given in the bottom row.

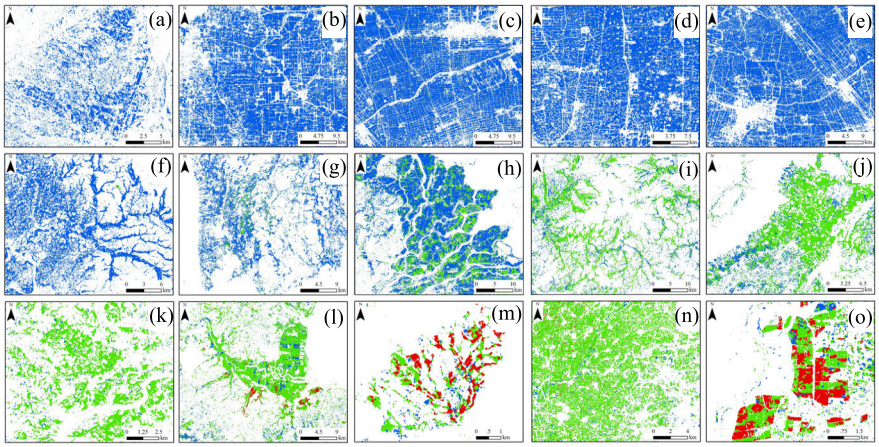


Fig. 13. Single (blue), double (green), and triple (red) rice-crop intensities at 15 sites.

At each site, the UA and PA are mostly higher than 75%, and the OA is near or higher than 80%. Since backscatter variations of different rice-crop intensities can sometimes be confused due to various interferences, the accuracies generally decrease as the regional multiple rice-crop suitability increases. The highest accuracy is achieved at site C, where only single-crop rice is cultivated on a flat terrain with uniform field shapes. The lowest accuracy is at site L because the single-, double-, and triple-crop rice fields and nonrice croplands are interlaced across the complex and undulating terrain. Still, when samples from all sites are combined, an OA of 82.26% is achieved (Table II). Thus, the assessment is satisfactory overall.

TABLE II
ACCURACY ASSESSMENT FOR ALL SITES

Predicted \ Actual	S	D	T	N	UA
S	1239	75	12	132	84.98%
D	66	765	30	63	82.79%
T	18	42	243	15	76.42%
N	180	60	15	1035	80.23%
PA	82.44%	81.21%	81.00%	83.13%	OA 82.26%

Agricultural census data derived from nine provinces are utilized to assess the accuracy of our results at a provincial scale (Table III). High consistency is achieved for single rice-crop intensity in Henan, Hunan, Jiangsu, Jiangxi, Shanghai, and Zhejiang, as well as for double rice-crop intensity in Hainan. However, the single rice-crop intensity in Sichuan and the double rice-crop intensity in Fujian are significantly underestimated, where the identified areas are less than half of the census values. The underestimation is likely because these provinces are characterized by mountainous terrain, with many rice fields located in areas with slope greater than 5°. These fields are not adequately preserved during the masking process. In contrast, rice fields in the high-consistency provinces are primarily situated in the major plains (e.g., Jianghan Plain,

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TABLE III

COMPARISON BETWEEN AGRICULTURAL CENSUS AND OUR RESULTS FOR SINGLE (S) AND DOUBLE (D) RICE-CROP INTENSITIES IN NINE PROVINCES

	Area (Km^2)			
Province	**S(census)**	**S(identified)**	**D(census)**	**D(identified)**
Henan	6,082	5,513	\	\
Hunan	14,761	15,931	\	\
Jiangsu	22,028	23,873	\	\
Jiangxi	9,458	10,914	\	\
Shanghai	1,041	972	\	\
Sichuan	18,663	8,013	\	\
Fujian	2,584	1,334	977	330
Zhejiang	4,452	3,945	896	532
Hainan	\	\	1,100	1,041

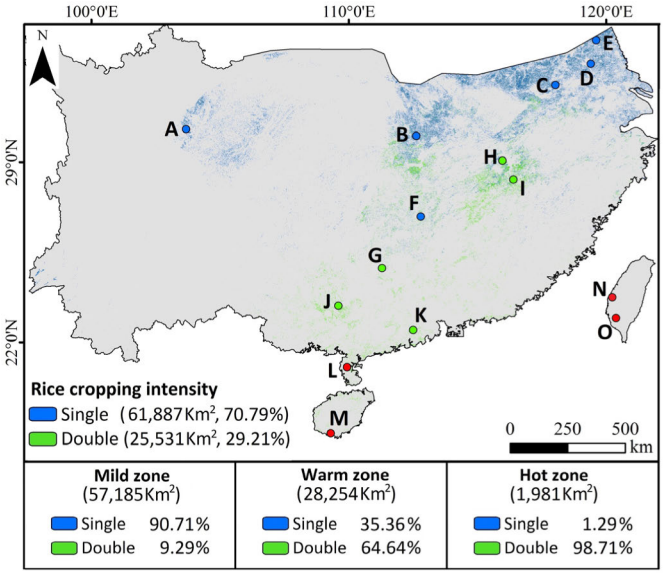


Fig. 14. Rice-crop intensities in Product2.

Ganjiang Plain, and Yangtze River Plain). Conducting regression analysis on our results against agricultural census data in the nine provinces, a coefficient of determination (R^2) of 0.82 and an RMSE of 3344 km^2 are obtained. By treating the single-intensity data in Sichuan and the double-intensity data in Fujian as outliers and removing them, an R^2 of 0.99 and an RMSE of 1010 km^2 can be achieved.

D. Product Comparison

Comparative analysis is conducted to assess the national-scale consistency of the rice-crop intensity identified in this study (hereafter called Product1) with other products, including a single [41] and double [42] rice-crop distribution dataset at 10-m resolution (Product2). Product2 identifies the single and double rice-crop fields using the Sentinel-1 data in 2020 and a time-weighted dynamic time warping (TWDTW) method. To compare Product1 and Product2 to the same geographical extent, we retain the low-altitude (<1000 m) and flat (<5°) fields in Product2, as shown in Fig. 14.

Product2 agrees overall spatially with Product1. However, the quantities and proportions of single and double rice-crop intensities differ significantly between the two products. Fields with a single rice-crop intensity in Product2 (61 887 km^2) accounts for only ~46% of that in Product1 (135 537 km^2), while the fields with a double rice-crop intensity in Product2 (25 531 km^2) are slightly more than that in Product1 (19 036 km^2). Single rice-crop fields in Product2 are significantly less than that in Product1, especially in the mild zone (48 626 km^2) of Southern China, where a single rice-crop intensity is temperature-appropriate, but a large number of rice fields are not identified. Product2 identifies around 6495 km^2 more fields with a double rice-crop intensity compared to Product1. These fields are mainly located in the mild zone (5313 km^2), where a double rice-crop practice is deemed unsuitable in this study.

TABLE IV

PIXELWISE CONSISTENCY BETWEEN THE TWO PRODUCTS FOR SINGLE (S), DOUBLE (D), AND TRIPLE (T) RICE-CROP INTENSITIES AND NONRICE (N) CROPLAND

		Product1			
		S	**D**	**T**	**N**
Product2	**S**	590,081,937 (36.6%)	8,106,787 (4.2%)	4,287 (0.2%)	172,797,815 (14.2%)
	D	136,313,536 (8.5%)	86,962,632 (45.3%)	1,626,976 (68.6%)	63,731,404 (5.2%)
	N	884,188,807 (54.9%)	97,068,889 (50.5%)	738,447 (31.2%)	978,446,168 (80.5%)
	Sum	1,610,584,280 (100%)	192,138,308 (100%)	2,369,710 (100%)	1,214,975,387 (100%)

Pixelwise statistics in Table IV provide a more intuitive understanding of the rice-crop intensity differences between the two products. The most significant distinction is that approximately 54.9% of single ones, 50.5% of double ones, and 31.2% of triple ones in Product1 are identified as nonrice cropland in Product2. This discrepancy is mainly attributed to two reasons. First, similar to other existing studies, Product2 detects the time-series backscatter features based on the assumption that the most significant trough occurs in the rice transplanting period. However, there is a widespread presence of atypical rice backscatter pattern, whose trough occurs in the reproductive period. The trough in the transplanting period has a short duration and is hardly detected. Second, Product2 identifies the rice growth cycles based on a typical phenology calendar without considering the diversity of rice phenology. Farmers may decide when to grow rice according to their own preferences and temperature conditions. In contrast, the time window for detecting rice growth cycles in Product1, determined based on the temperature suitability, is generally wider than that in Product2. As a result, fields following the atypical phenology calendar can be identified in Product1.

Although only 45.3% of pixels with a double rice-crop intensity in Product1 are identified as double ones in Product2, Product2 recognizes more double rice-crop fields. This can be attributed to about 8.5% of pixels with a single rice-crop intensity in Product1 being identified as double ones in Product2. While the proportion is not high, it significantly expands the double rice-crop area due to the large area of single ones. These fields are situated in areas where farmers follow a

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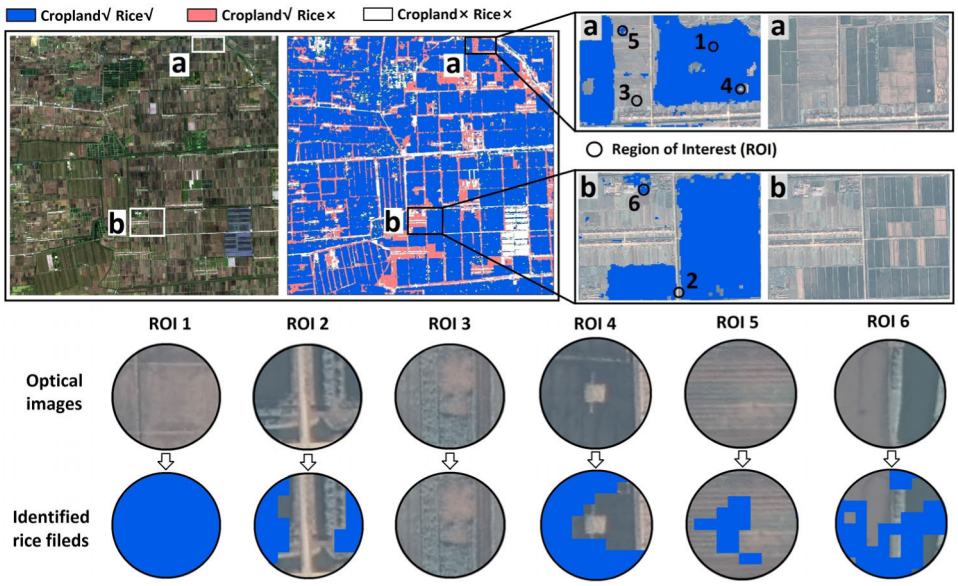


Fig. 15. Rice-crop intensity identified at site B. Regions of interest (ROI) are circled and numbered. ROI 1 shows rice fields. ROI 2 shows ridges, channels, and trees near rice fields. ROI 3 shows orchards and bare land. ROI 4 shows an electric tower within a rice field. ROI 5 shows a lettuce field. ROI 6 shows a peanut field near a pond. Rice fields are accurately identified in ROI 1–3, while errors occur in ROI 4–6.

TABLE V
ACCURACY ASSESSMENT OF PRODUCT2 BASED ON SAMPLE FIELDS

		Actual			
		S	D	N	UA
Predicted	S	642	63	117	78.10%
	D	189	573	129	64.31%
	N	540	69	942	60.74%
	PA	46.83%	81.28%	79.29%	OA 66.08%

seasonal pattern of rotating between rice and nonrice crops. This practice helps maintain soil fertility and enhance crop yield, especially under suboptimal temperatures.

Product2 achieves an OA of 66.08% based on our sample fields, as shown in Table V. About half of the sample fields with single rice-crop intensity are misclassified as nonrice cropland, leading to a PA of 46.83% for single rice-crop intensity and a UA of 60.74% for nonrice cropland. In addition, many nonrice fields and fields with single rice-crop intensity are classified as double ones, resulting in a UA of 64.31% for double rice-crop intensity, indicating seasonal rotated crops are identified as rice. The sample validation in Table V generally aligns with the pixelwise statistics presented in Table IV.

V. DISCUSSION

A. Error and Uncertainty Analysis

To better understand the mapping accuracy, we select a subset of fields at sites B, H, and L as examples. Fig. 15 shows the subset of fields at site B, where the terrain is suitable for rice cropping, but the temperature is only suitable for a single rice-crop intensity. Since the fields are identified based on the cropland area in the ESA WorldCover product, the cropland layer is placed underneath the rice layer. Then, the blue areas indicate rice fields, the pink areas are for nonrice cropland, and the blank areas indicate noncropland. The results demonstrate the successful discrimination of nonrice cropland (pink polygonal features) from rice fields. Additionally, the results effectively detect and highlight the ridges (pink linear features), often misclassified as cropland in the ESA WorldCover product.

Multiple troughs may occur annually in a rice field, but only one deep trough occurs during the temperature-appropriate period for rice cropping, as shown in ROI 1 in Fig. 16. The ridges typically stand 20–50 cm taller than the surrounding cropland, and ridge backscatter stay consistently high (ROI 2 in Fig. 16) due to the dihedral effect formed by the field-ridge structure. Dry-land crops such as corn, canola, and wheat are usually planted in the nonrice cropland. Although their growth processes may also cause backscatter fluctuations, the troughs typically occur in periods unsuitable for rice cropping or have relatively high backscatter values ($\geq$0.02) on the trough days due to the rough soil surface (ROI 3 in Fig. 16).

A few misclassifications between rice fields and nonrice cropland can occur, that is, some isolated rice pixels appear on the nonrice cropland, and some pixels in the rice fields are missing, as shown in ROI 4 and 5 in Fig. 15. Rice fields can be wrongly identified as nonrice cropland when nonrice objects such as graves, transmission, or communication towers are present in the rice fields. In these regions, rice backscatter fluctuations are influenced by the backscatter from adjacent nonrice objects, resulting in the backscatter values sometimes higher than 0.02 on the trough days (ROI 4 in Fig. 16). On the other hand, the nonrice cropland area can be wrongly identified as rice fields when the areas have smooth soil surfaces or rainwater puddles, during which the backscatter variations may be lower than 0.02 (ROI 5 in Fig. 16). In addition, the nonrice cropland adjacent to water bodies, such as ponds, rivers, and lakes, may be misclassified as rice fields due to backscatter declines caused by the land flooding in the rainy season (ROI 6 in Fig. 16).

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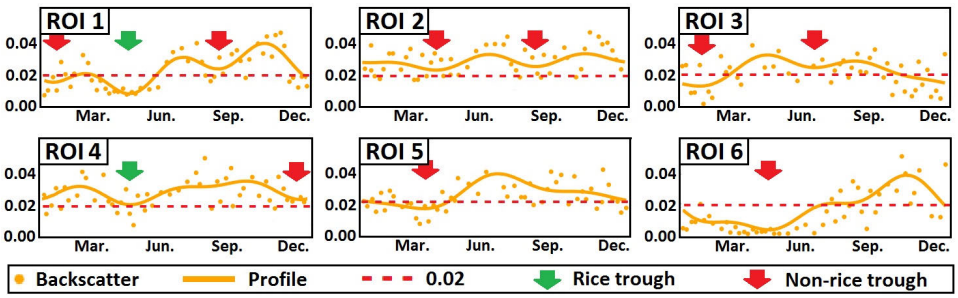


Fig. 16. Backscatter variations from rice fields in ROI 1 and 4, ridges in ROI 2, bare land in ROI 3, lettuce fields in ROI 5, and peanut fields in ROI 6 at site B. The green arrow indicates a rice trough, while the red arrow indicates a nonrice trough.

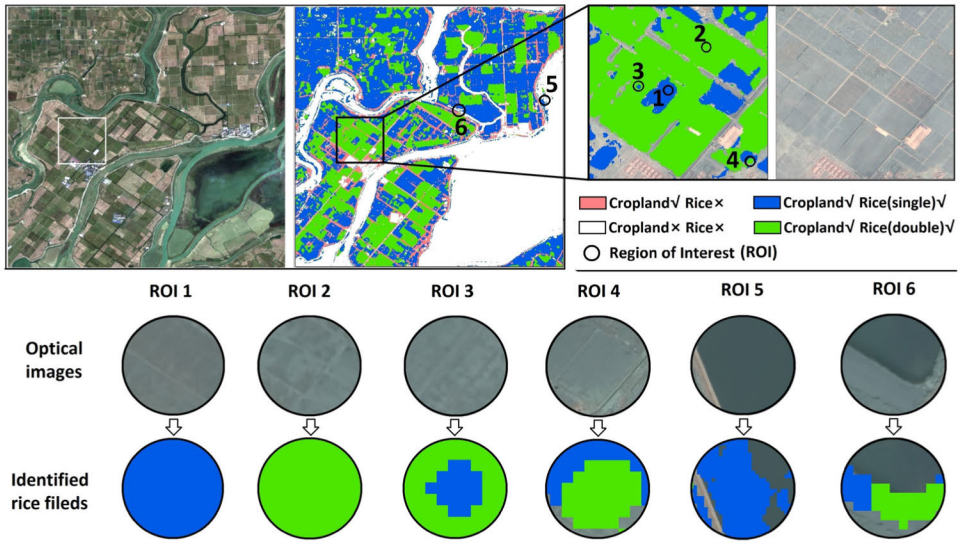


Fig. 17. Rice-crop intensity identified at site H. ROI 1 and 2 are fields correctly identified as single and double intensities. ROI 3 shows double rice-crop area misclassified as single intensity (blue pixels) due to shallow water during the second cycle's transplanting. ROI 4 shows single rice-crop area misclassified as double intensity (green pixels) due to rainfall waterlogging after rice harvesting. ROI 5 shows riverside noncropland misidentified as single intensity (blue pixels), and ROI 6 shows riverside single rice-crop area misidentified as double intensity (green pixels) due to rainy season flooding.

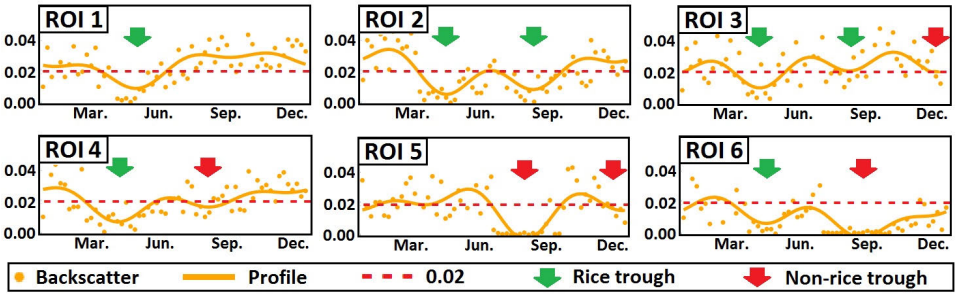


Fig. 18. Backscatter variations from single rice-crop areas in ROI 1, 4, and 6, double rice-crop areas in ROI 2 and 3, and noncropland in ROI 5 at site H.

Fig. 17 shows a subset of fields at site H, where the temperature is suitable for a double rice-crop intensity. The results effectively differentiate between the nonrice cropland, ridges, and single and double rice-crop fields.

One and two deep troughs are observed during the temperature-appropriate period of the single (ROI 1 in Fig. 18) and double (ROI 2 in Fig. 18) rice-crop intensities, respectively. However, there is also a possibility of misclassifications when the backscatter variations approach 0.02 on the trough dates, resulting in missed detection (trough near September for ROI 3 in Fig. 18) or false detection (trough near September for ROI 4 in Fig. 18) of rice troughs. Site H is located near the Poyang Lake, and some cropland areas near the water may get flooded in the rainy season, bringing about false detection (trough near September for ROI 5 and 6 in Fig. 18) of rice troughs and consequent overestimation of rice-crop intensity. The weak backscatter from flooded fields in ROI 5 occurs from early July to late August and in ROI 6 from mid-July to mid-September. The approximately half-month delay in ROI 6 compared to ROI 5 is attributed to ROI 6 being farther from the lake, with obstacles such as fields and ridges impeding the flooding and draining processes.

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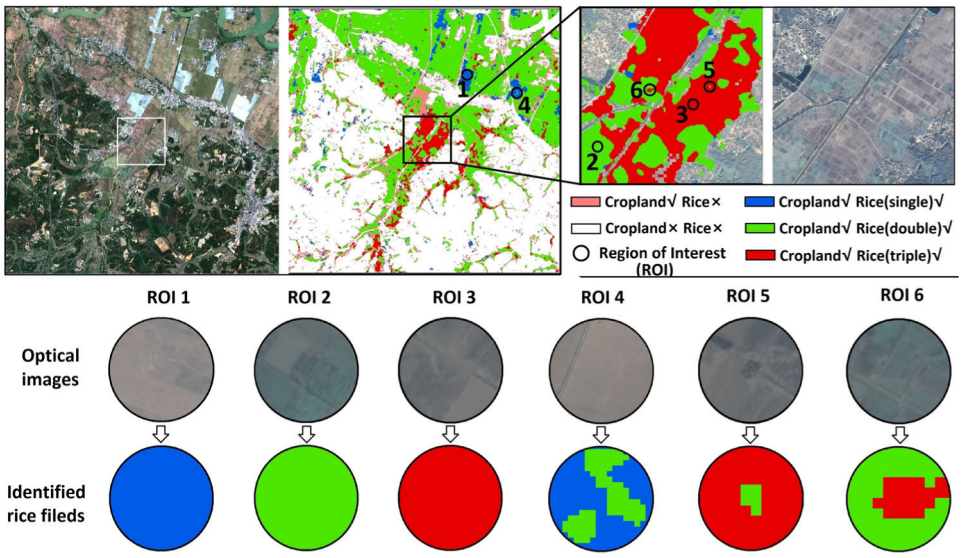


Fig. 19. Rice-crop intensity identified at site L. ROI 1–3 are fields correctly identified as single, double, and triple intensities, respectively. ROI 4 shows single rice-crop area misclassified as double intensity (green pixels), and ROI 6 shows double rice-crop area misclassified as triple intensity (red pixels) due to rainfall waterlogging after rice harvesting. ROI 5 shows triple rice-crop area misclassified as double intensity (green pixels) due to shallow water during the last cycle's transplanting.

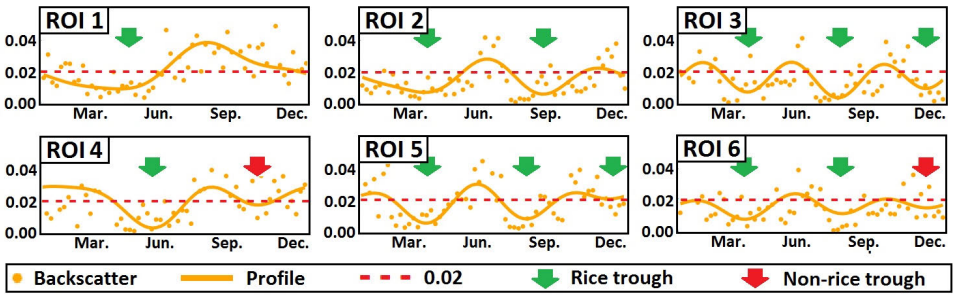


Fig. 20. Backscatter variations from single rice-crop areas in ROI 1 and 4, double rice-crop areas in ROI 2 and 6, and triple rice-crop areas in ROI 3 and 5 at site L.

Fig. 19 shows a subset of fields at site L, where the temperature suits a tripe rice-crop intensity. However, the dominant rice-crop practice in this area is double because of the time- and labor-intensive for the triple rice-crop intensity.

The single, double, and triple rice-crop intensities exhibited one (ROI 1 in Fig. 20), two (ROI 2 in Fig. 20), and three (ROI 3 in Fig. 20) deep troughs, respectively. However, the tropical monsoon climate at site L produces frequent rainfall and high soil moisture in the cropland. The high soil moisture can cause low backscatter values (<0.02) during nonrice land processes, leading to false detection (troughs near December for ROI 4 and 6 in Fig. 20) of rice troughs. On the other hand, the third rice-crop cycle is usually transplanted in December with lower temperature and sunlight supply than the previous cycles. As a result, less water is kept in the fields during the third rice-crop cycle, and the rice trough is sometimes higher than 0.02 (trough near December for ROI 5 in Fig. 20), leading to missed detection. Although some errors exist at sites B, H, and L, the overall performance is still satisfactory as the rice-crop intensities in most fields are correctly identified.

The total area of rice fields identified in this study is 154 831 km^2. The cropland area in the ESA WorldCover product is 256 097 km^2 after excluding the fields unsuitable (where the suitability mask is gray) for rice cropping. Then, the rice planting areas are ~61% of the cropland area. Considering the widespread nonrice cropland in Southern China and the OA of 80.7% achieved by the ESA WorldCover product in Asia, one should consider the proportion of ~61% reasonable.

Compared with agricultural census data, the identified rice planting areas in some provinces with mountainous terrain (e.g., Sichuan and Fujian) are underestimated. This underestimation occurs due to the omission of a portion of rice planted in nonflat (slope $>$ 5°) cropland during terrain masking. An example of the mountainous scenario in Sichuan Province is shown in Fig. 21.

The nonflat cropland is unsuitable for rice cultivation due to the challenges associated with irrigating and restoring water, and thus, it is primarily dedicated to cultivating nonrice crops such as corn, potatoes, and vegetables. The backscatter variations of these nonrice crops sometimes closely resemble those of rice. Effective differentiation between them (e.g., ROI B in Fig. 21) can be achieved using the slope mask. However, rice may be cultivated on nonflat cropland in the form of terraces or on flat cropland in narrow valleys to ensure food production. Rice terraces with a slope $>$ 5° are omitted (e.g., ROI C in Fig. 21) during slope masking. On the other hand,

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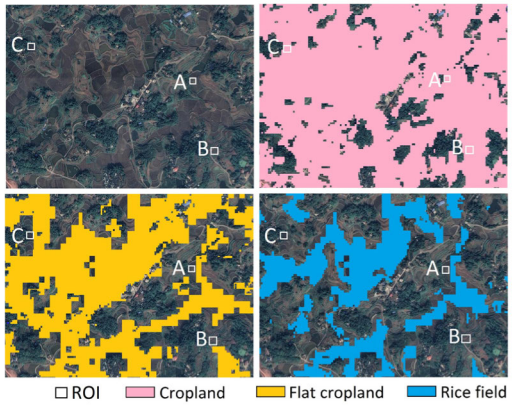


Fig. 21. Example of the mountainous scenario in Sichuan Province, including cropland and flat (slope $\leq 5°$) cropland delineated by the ESA WorldCover product and NASADEM data, and rice fields identified in this study.

TABLE VI
PROPORTION OF NONFLAT FIELDS IN CROPLAND AREA

Province	Proportion	Province	Proportion
Shanghai	97.79%	**Jiangxi**	81.31%
Jiangsu	96.88%	**Hunan**	73.65%
Henan	89.28%	**Fujian**	65.56%
Hainan	87.70%	**Sichuan**	49.25%
Zhejiang	84.75%		

rice planted near the edges of valleys (e.g., ROI A in Fig. 21) may be excluded because the cropland is located in transitional zones between flat and nonflat areas, where the resampled NASADEM slope sometimes exceeds 5°. These two types of omissions lead to the underestimation of the rice planting area.

Provincial-scale statistics are conducted (Table VI) to measure the proportion of nonflat cropland in all cropland areas, utilizing the ESA WorldCover product and the slope mask. Provinces with a higher proportion of nonflat cropland, such as Sichuan and Fujian, tend to underestimate the rice planting area more.

B. Impact of Temperature Information

This study makes an important contribution by utilizing regional temperature information in combination with an understanding of rice plant physiology, effectively discerning the backscatter characteristics of rice growth and nonrice processes under complex and diverse agricultural practices in Southern China. To visually illustrate the impact of temperature information on rice growth cycle identification, trough detection results before implementing temperature restriction and overestimation correction are mapped in Fig. 22.

Harmonic analysis is conducted on the time-series backscatter within the suitable rice-crop areas (mild, warm, and hot zones). Then, deep troughs are detected and counted before temperature restriction, as shown in Fig. 22(a). The double-trough fields dominate (76.05%) all suitable areas, while the single- and triple-trough fields account for 22.14% and 1.81%, respectively. Single- and double-trough fields are generally mixed, indicating diverse farming practices. Troughs are recognized correctly, but nonrice troughs cannot be distinguished from rice ones. The proportions contradict the actual rice cultivation practices if all troughs correspond to rice growth cycles, as the double-trough fields dominate even in the mild zone, where the temperatures only support a single rice-crop intensity.

After conducting the temperature restriction, only the troughs that meet the phenological temperature requirements are retained, as shown in Fig. 22(b). The single-trough fields increase to 80.85%, while the double- and triple-trough fields decrease to 18.88% and 0.27%, respectively. The total area (154 831 km^2) of rice fields in Fig. 22(b) is less than that (165 890 km^2) in Fig. 22(a), indicating that deep backscatter troughs in some fields occur in the cold period unsuitable for rice growth. Thus, these fields are corrected as nonrice fields. Remarkable changes occur in the mild and warm zones, where about half of the double-trough fields are corrected as single-trough ones.

However, double-trough fields still appear in the northwest plain near site C, where the nonrice crops (e.g., spring wheat) with low sunlight and temperature requirements are planted before rice cultivation to maximize the field's productivity. The tall-stem spring wheat and the flat underlying surface cause strong microwave attenuation, creating nonnegligible troughs similar to rice troughs. After the overestimation correction, rice-crop intensity in Southern China is finally mapped in Fig. 12(b). About 10 191 km^2 of the double-trough fields are corrected to the single rice-crop intensity, and 159 km^2 of the triple-trough fields are corrected to the double rice-crop intensity.

Multiple rice-crop intensities indicate the multiplication of rice planting areas over time. Specifically, a double rice-crop intensity doubles the planting area, and a triple intensity triples it. As a result, in 2020, the equivalent rice planting areas are 174 386 km^2. Cropland in the suitable rice-crop areas is 256 097 km^2, characterized by flat terrain and low altitude, with a monsoon climate that ensures sufficient water and temperature for at least one rice growth cycle. In addition, irrigation systems ensure a stable water supply to rice paddies during the growth cycle if rainfall is inadequate. Therefore, the cropland area in warm and hot zones is potentially suitable for multiple rice-crop intensities. Thus, the maximum potential equivalent rice planting area in 2020 is 361 123 km^2. However, the actual equivalent rice planting area (174 386 km^2) only accounts for 48.29% of the maximum potential (361 123 km^2), which can be attributed to farmers' considerations of maximizing the economic benefits of planting different crop types and optimizing their investment of time and labor in the cropping process. Multiple rice-crop intensities require harvesting the former cycle and transplantation of the latter cycle within a short time frame of approximately ten days. Due to the ongoing urbanization process, China is facing a shortage of rural labor, leading to the double or triple rice-crop intensity no longer being the preferred rice cropping scheme.

C. Data Dependence in the Method

The dependence of the proposed method on SAR data is analyzed from the perspectives of the length, revisit period,

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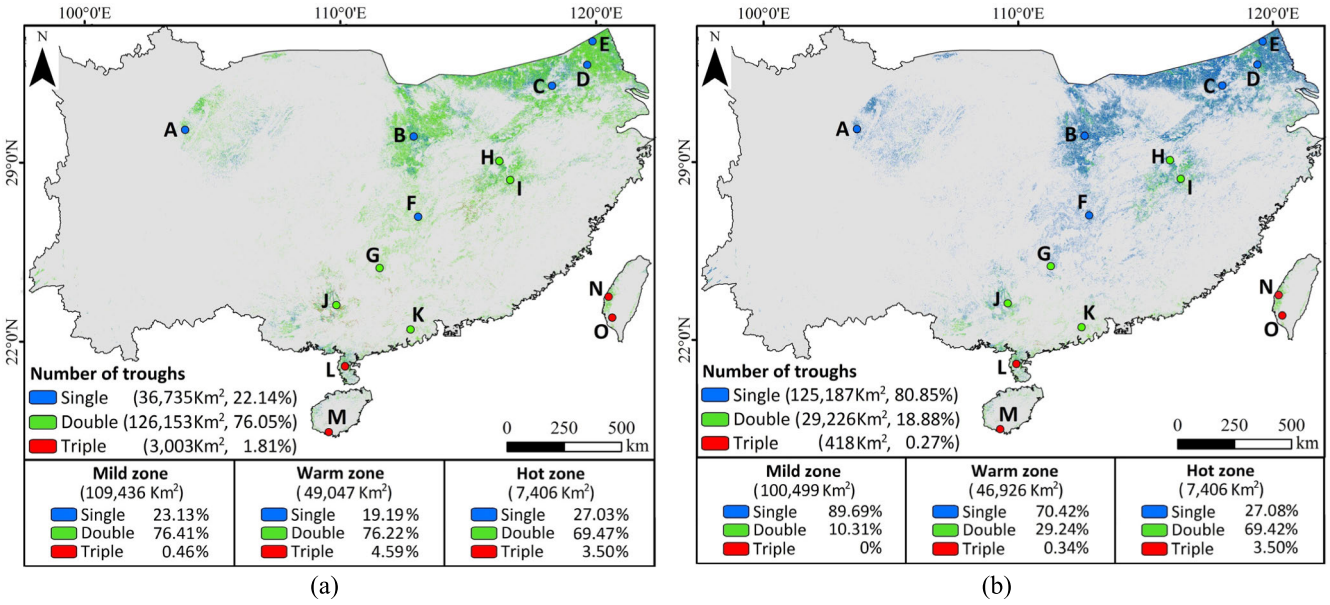


Fig. 22. (a) Number of troughs before temperature restriction and overestimation correction. (b) Number of troughs after temperature restriction but before overestimation correction.

and radar incidence angle of the time-series observations. The method proposed in this study requires time-series data with a length sufficient to monitor all rice growth cycles in the study area. Theoretically, identifying the single, double, or triple rice-crop intensity requires data within a time window of at least 120, 240, or 360 days, respectively. However, the unavailability of prior phenological information makes it impractical to determine these time windows at a national scale. Therefore, in this study, the length of time-series data used for each location is the entire year (366 days).

The revisit period determines the temporal resolution of SAR observations for monitoring rice growth process. The key to identifying rice-crop intensity lies in detecting weak backscatter (<0.02) during the trough period, which typically lasts about one month. Therefore, it is imperative to ensure at least one SAR observation conducted monthly. Assuming a constant revisit period, at least 12 SAR images are required. Furthermore, considering the potential occurrence of speckle noise and unexpected land change events, it is recommended to conduct more than two observations per month to mitigate the impact of occasional disturbances. This means obtaining 24 or more images annually, which can enhance the reliability of the time-series analysis. The revisit period for Sentinel-1A/B is 12 days in Southern China, adequately meeting the requirements. In this study, both Sentinel-1A&B are used whenever available. Sentinel-1B ceases operations starting from 2022. However, the unavailability of Sentinel-1B data would not affect our scheme. In fact, Sentinel-1B data are available for only part of Southern China in 2020, as shown in Fig. 23.

In areas where the Sentinel-1A/B sensor can capture images (e.g., strip1 in Fig. 23), approximately 30 observations are routinely conducted, with a revisit period of 12 days. In areas where both A&B sensors can acquire images (i.e., the region outlined by the black polygons in Fig. 23), about 60 SAR observations are routinely performed, with a revisit period

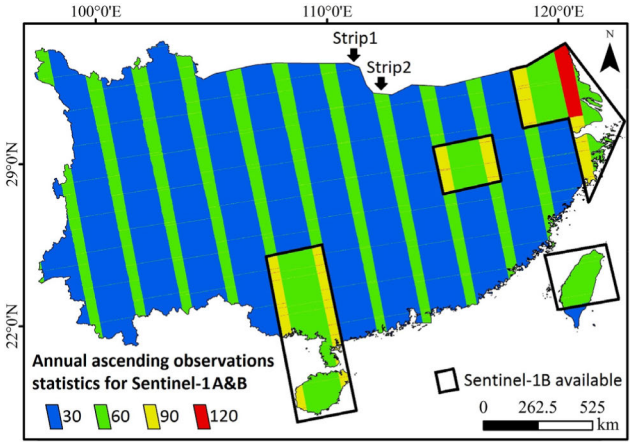
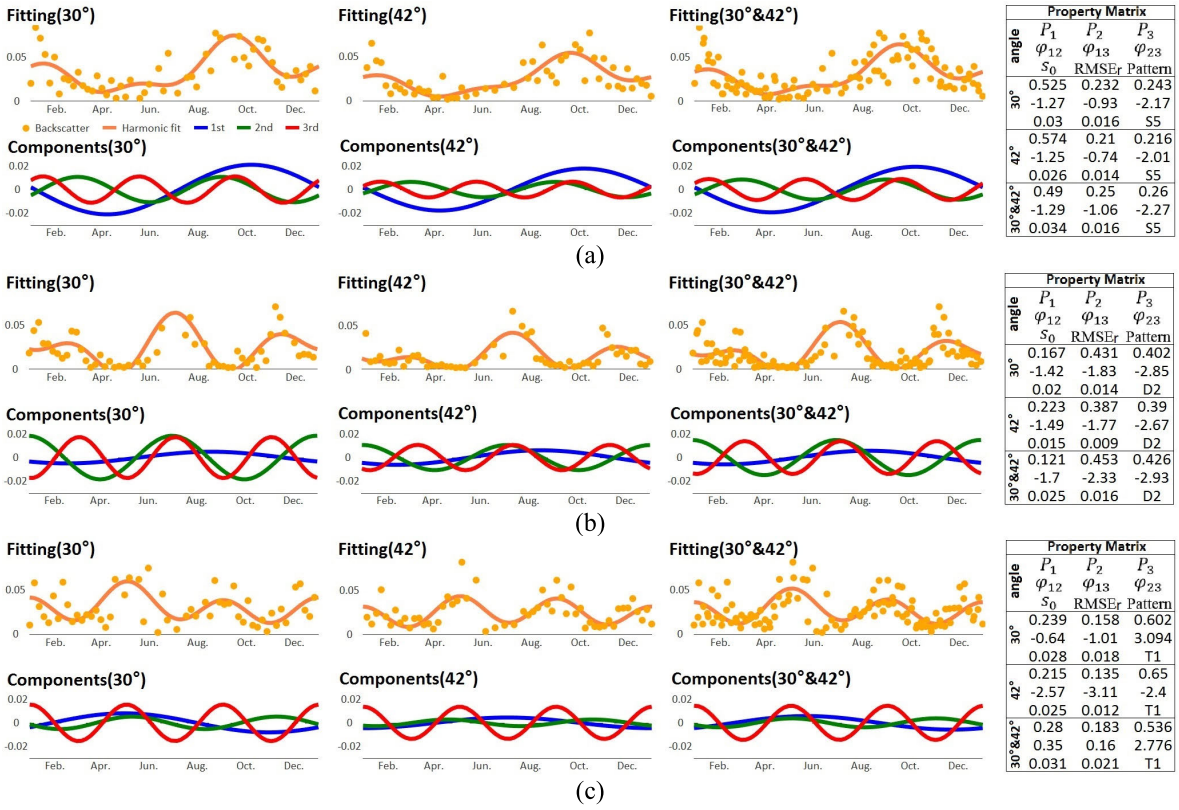


Fig. 23. Number of Sentinel-1A&B ascending observations for Southern China in 2020.

of six days. In areas with overlapping swaths (e.g., strip2 in Fig. 23) from neighboring orbits, the number of observations increases compared to individual orbits, with a revisit period ranging from 3 to 6 days. Thus, the observation frequency is sufficiently high for detecting backscatter troughs within multiple rice growth cycles.

The radar incidence angle determines the observed structural properties of the rice plants and underlying surface. Variations in the radar incidence angle can lead to changes in the backscatter characteristics of rice fields, i.e., the backscatter intensity at specific imaging time. However, the proposed method focuses on the trends of time series. The stable orbits of Sentinel-1 ensure that most fields are observed at a constant incidence angle throughout the year, maintaining the stability of the temporal trends. Only rice fields located in the swath overlapping areas are observed at varying incidence angles. To explore the influence of the incidence angle on time-series trends and trough features, three samples (Fig. 24) are derived

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Property Matrix

angle	P_1	P_2	P_3
	φ_{12}	φ_{13}	φ_{23}
	s_0	RMSEr	Pattern
30°	0.525	0.232	0.243
	-1.27	-0.93	-2.17
	0.03	0.016	S5
42°	0.574	0.21	0.216
	-1.25	-0.74	-2.01
	0.026	0.014	S5
30°&42°	0.49	0.25	0.26
	-1.29	-1.06	-2.27
	0.034	0.016	S5

Property Matrix

angle	P_1	P_2	P_3
	φ_{12}	φ_{13}	φ_{23}
	s_0	RMSEr	Pattern
30°	0.167	0.431	0.402
	-1.42	-1.83	-2.85
	0.02	0.014	D2
42°	0.223	0.387	0.39
	-1.49	-1.77	-2.67
	0.015	0.009	D2
30°&42°	0.121	0.453	0.426
	-1.7	-2.33	-2.93
	0.025	0.016	D2

Property Matrix

angle	P_1	P_2	P_3
	φ_{12}	φ_{13}	φ_{23}
	s_0	RMSEr	Pattern
30°	0.239	0.158	0.602
	-0.64	-1.01	3.094
	0.028	0.018	T1
42°	0.215	0.135	0.65
	-2.57	-3.11	-2.4
	0.025	0.012	T1
30°&42°	0.28	0.183	0.536
	0.35	0.16	2.776
	0.031	0.021	T1

Fig. 24. Samples of (a) single, (b) double, and (c) triple trough patterns, illustrating the impact of radar incidence angle variation in the swath overlapping area. For each sample, the first column represents observations with the radar incidence angle of 30°, the second column with the angle of 42°, the third column with the angle alternating between 30° and 42°, and the last column is the corresponding harmonic features.

from the swath overlapping area, where rice backscatter is acquired alternately at the near and far range of the SAR imaging on different orbits, with small (30°) and large (42°) radar incidence angles alternating. While variations in radar incidence angle could impact the backscatter characteristics of rice fields, they have minimal influence on the harmonic parameters and profile patterns of the time series due to the consistent temporal trend. The timing and backscatter intensity of the troughs remain unchanged. Hence, this study does not take into account the impact of the radar incidence angle on mapping results.

D. Significances of the Proposed Scheme

The method proposed in this study extends the SAR-based rice mapping capability in the time domain. Recent SAR-based rice mapping researches, such as [14] and [43], have gained widespread recognition. Both studies share a common strategy of identifying the low backscatter during rice transplanting, as well as the backscatter increase after rice transplanting. The difference is that Zhan et al. [14] establish a series of thresholds to restrict the linear slope of backscatter variations before and after rice transplanting. This is designed to identify the distinctive V-shaped backscatter dynamics. On the other hand, Xu et al. [43] calculate an SAR-based paddy rice index (SPRI) based on local minimal and maximal values of the time series. A threshold of SPRI is then set to identify paddy fields. In theory, the two methods can more accurately recognize the morphology of rice backscatter troughs compared to our method. However, Zhan et al. [14] method is applicable only to regions with available prior phenological data, relying on crop calendars to determine the time window of field flooding. Xu et al. [43] method requires pointwise differencing of smoothed time series to search for inflection points, with specific restrictions to exclude pseudo-features caused by rainfall events. Both studies acknowledged limitations in identifying atypical backscatter variations of late rice, i.e., short duration of flooding signal during transplanting and extended backscatter trough during the reproductive period. In contrast, our method effectively extracts features through harmonic analysis and identifies atypical rice backscatter variations without using prior phenological information.

Thus, four significances are argued for the scheme proposed in this study. First, distinguishing backscatter variations under typical and atypical temperatures and recognizing the reliable trough feature. Rice backscatter troughs may occur in the transplanting or reproductive period, while crests may occur in the vegetative and mature periods or even be undetectable within a growth cycle. Here, a unique dominant trough is always found within a rice growth cycle.

Second, the harmonic analysis provides a more refined spatial homogeneity at the field scale, removing time-series interference by identifying the dominant periodicities of potential cyclical land processes. The harmonic decomposition method eliminates the necessity of specifying a filtering window or assuming a profile shape. It can efficiently integrate long and dense time series on a large scale, describing

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smoothed data with only seven cosine parameters. Fluctuation profiles and troughs are easily identified using an efficient harmonic feature recognition method, thus obviating the need for analyzing local trends or detecting inflection points.

Third, temperature restrictions are used instead of the traditional previous phenological information. Southern China covers about 2.6 million km^2 with diverse social and natural environments. The rice phenology calendar differs at region and field levels. Here, potential rice growth cycles are identified by analyzing the temperature supply. The temperature-appropriate period for detecting rice troughs is determined, considering the temperature requirements of rice plants during different phenological periods.

Finally, the overestimation of rice-crop intensity is corrected appropriately based on the regional multiple rice-crop suitability. The existing SAR-based and optical-based cropping intensity products are significantly uncertain in differentiating the rice growth and nonrice processes. Here, the overestimation is identified when the number of suspected rice grow cycles exceeds the regional maximum suitability. Features related to nonrice processes are recognized according to their unfavorable temperatures.

VI. Conclusion

We developed a novel scheme implemented on the GEE platform to identify rice-crop intensity at high resolution and large scale based on harmonic analysis of time-series Sentinel-1 VH backscatter, coupling with regional temperature analysis. The spatial distribution of single, double, and triple rice-crop intensities is mapped in 2020. Accuracy assessments at a regional scale and comparative analysis at a provincial and national scale confirm the reliability of our product. The area of rice fields in Southern China is 154 832 km^2 in 2020. The single rice-crop intensity is the dominant practice, comprising 87.54% of the total, and is primarily distributed within the subtropical zones of Southern China. The double rice-crop intensity (12.29%) is distributed in the central and southern subtropical zones. The triple rice-crop intensity (0.17%) is located in the southernmost tropical zone. The developed method has effectively distinguished the rice growth cycles and nonrice land processes without using previous phenological information. The methodology should further support rice yield prediction and agricultural/ecological policy-making beyond the borders of China.

Appendix

Phase-shift (φ_{ij}) analysis of nonnegligible components is conducted with a step increment of $\pi/4$. Although φ_{ij} ranges from $-\pi$ to π, only the synthesis profiles with a positive φ_{ij} are shown because the profiles with a negative φ_{ij} is symmetrical to that with a positive φ_{ij}. T_P in Figs. 25–28 denotes the nonnegligible troughs on the profiles.

As shown in Fig. 25, when only $s_1[t]$ and $s_2[t]$ are nonnegligible, the synthesis profiles always have two troughs. When the depth of a trough is incomparable to another within a specific φ_{ij} range, the shallower trough is negligible, and the profile belongs to a single-trough pattern. Thus, the profile belongs to a single-trough pattern (pattern S3) when $|\varphi_{12}| > \pi/4$, and belongs to a double-trough pattern (pattern D3) when $|\varphi_{12}| \le \pi/4$.

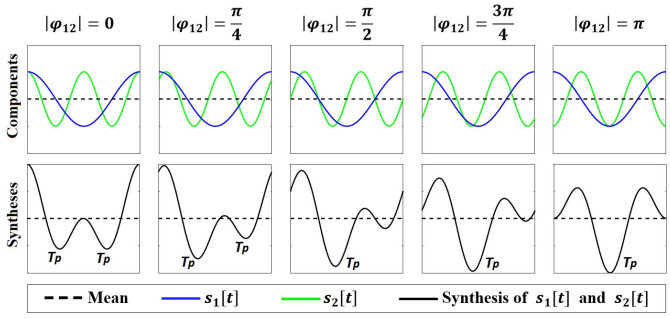


Fig. 25. Phase-shift analysis between $s_1[t]$ and $s_2[t]$.

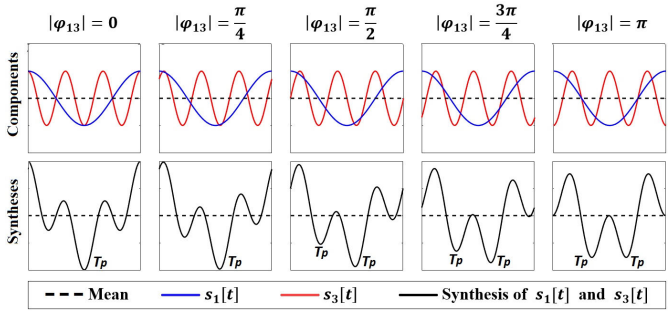


Fig. 26. Phase-shift analysis between $s_1[t]$ and $s_3[t]$.

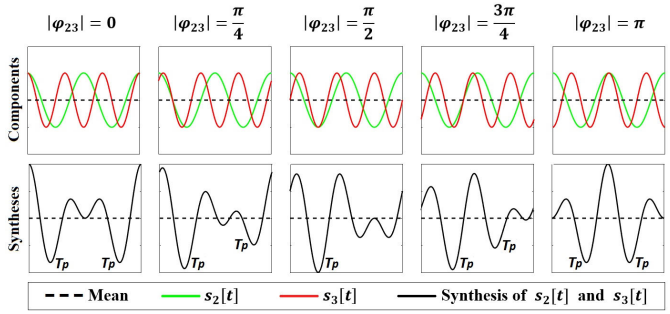


Fig. 27. Phase-shift analysis between $s_2[t]$ and $s_3[t]$.

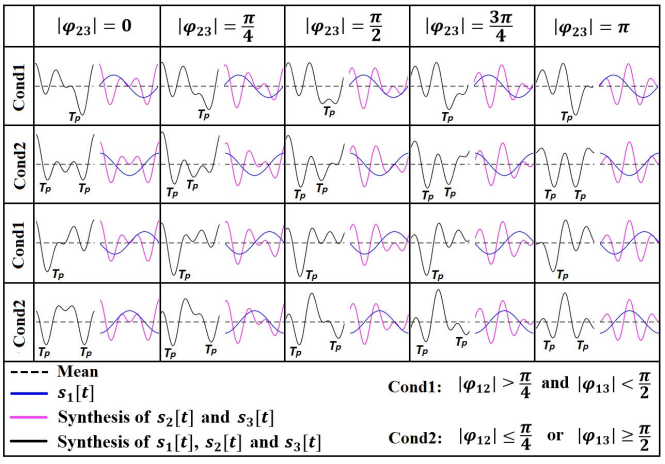


Fig. 28. Phase-shift analysis among $s_1[t]$, $s_2[t]$, and $s_3[t]$.

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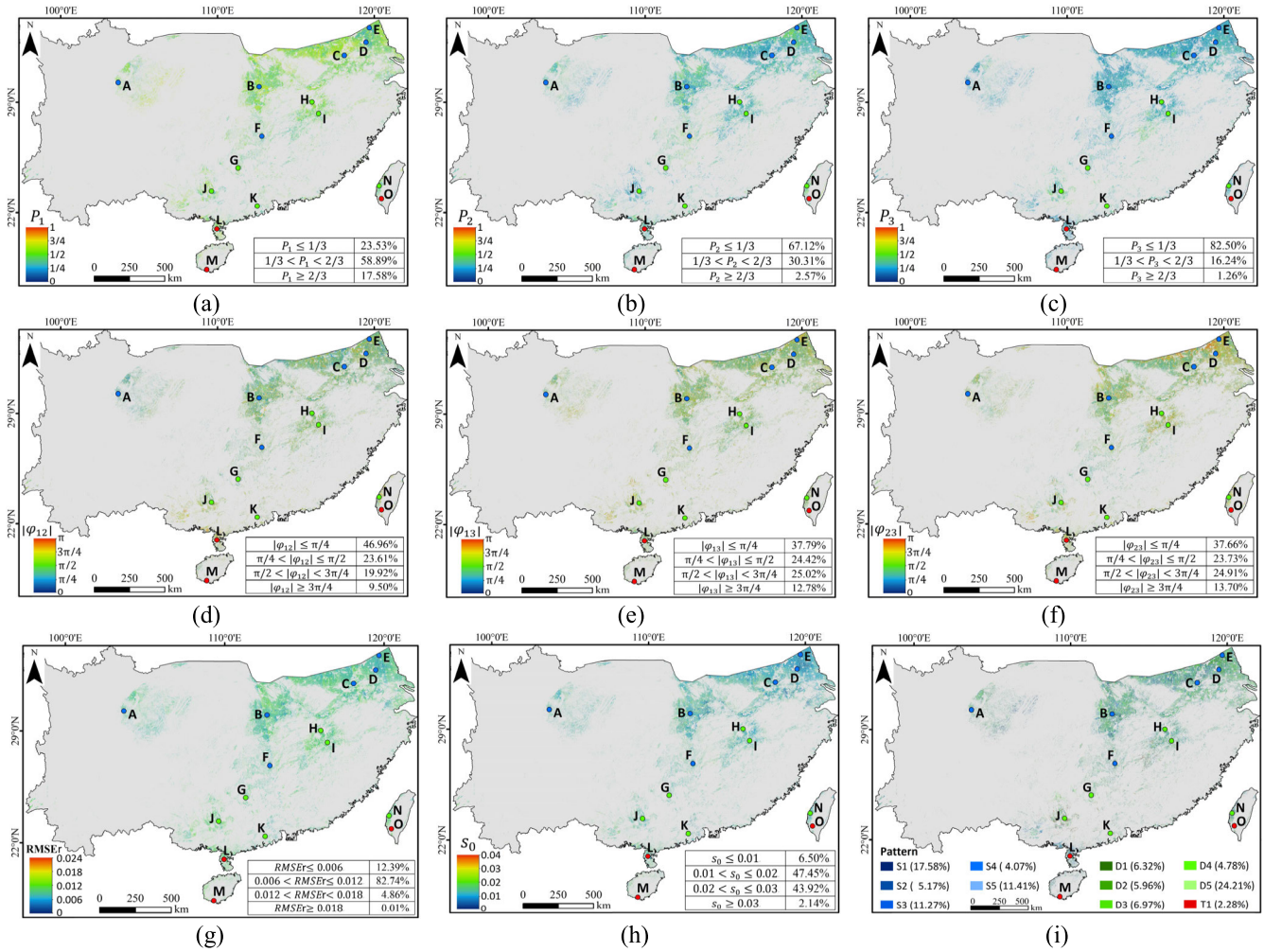


Fig. 29. Spatial distributions of (a) P_1, (b) P_2, (c) P_3, (d) φ_{12}, (e) φ_{13}, (f) φ_{23}, (g) RMSE$_r$, (h) s_0, and (i) profile patterns.

Similarly, as shown in Fig. 26, when only $s_1[t]$ and $s_3[t]$ are nonnegligible, the synthesis profiles always have three troughs. When the depth of a trough is incomparable to the deepest one within a specific φ_{ij} range, the shallower trough is negligible. Thus, the synthesis profile belongs to a single-trough pattern (pattern S4) when $|\varphi_{13}| < \pi/2$, and belongs to a double-trough (pattern D4) pattern when $|\varphi_{13}| \geq \pi/2$.

As shown in Fig. 27, when only $s_2[t]$ and $s_3[t]$ are nonnegligible, the synthesis profiles always have three troughs. The profile belongs to a single-trough pattern (pattern S2) when $\pi/4 < |\varphi_{23}| < 3\pi/4$ and belongs to a double-trough pattern (pattern D2) when $|\varphi_{23}| \leq \pi/4$ or $|\varphi_{23}| \geq 3\pi/4$.

As shown in Fig. 28, $s_1[t]$, $s_2[t]$, and $s_3[t]$ are all nonnegligible, and their synthesis profiles can be considered the synthesis of $s_1[t]$ with the synthesis profile (Fig. 27) of $s_2[t]$ and $s_3[t]$. The profiles of the three components belong to a single-trough pattern (pattern S5) when $|\varphi_{12}| > \pi/4$ and $|\varphi_{13}| < \pi/2$, and belong to a double-trough pattern (pattern D5) when $|\varphi_{12}| \leq \pi/4$ or $|\varphi_{13}| \geq \pi/2$.

The spatial distributions of harmonic parameters (P_i, φ_{ij}, s_0, and RMSE$_r$) and profile patterns within the rice planting area are shown in Fig. 29. $s_1[t]$ dominates ($P_1 \geq 2/3$) only 17.58% of rice fields, resulting in the pattern S1 constituting 17.58% of the total. They are mainly rain-fed fields in the mountain valleys, where rainwater accumulates during the rice growing season. The soil is usually rough and dry in other seasons.

$s_1[t]$ is negligible ($P_1 \leq 1/3$) in 23.53% of rice fields. Among them, fields with $P_2 \geq 2P_3$ account for 26.84%, resulting in the pattern D1 constituting 6.32% of the total. Fields with $P_3 \geq 2P_2$ account for 9.68%, leading to the pattern T1 constituting 2.28% of the total. Fields with comparable $s_2[t]$ and $s_3[t]$ (i.e., $P_2 < 2P_3$ and $P_3 < 2P_2$) account for 47.26%, consisting of fields with synchronized (21.95%) and unsynchronized (25.31%) second- and third-order troughs, resulting in the proportion of 5.17% and 5.96% of the total for patterns S2 and D2.

$s_1[t]$ is nonnegligible but not dominant ($1/3 < P_1 < 2/3$) in most (58.89%) fields. Among them, fields with comparable $s_2[t]$ and $s_3[t]$ account for 60.49%, consisting of fields with synchronized (19.37%) and unsynchronized (41.12%) third-order troughs, resulting in the proportion of 11.41% and 24.21% of the total for the patterns S5 and D5. Fields with $P_2 \geq 2P_3$ account for 30.96%, consisting of fields with synchronized (19.13%) and unsynchronized (11.83%) first- and second-order troughs, resulting in the proportion of 11.27% and 6.97% of the total for the patterns S3 and D3. Fields

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with $P_3 \geq 2P_2$ account for 15.03%, consisting of fields with synchronized (6.92%) and unsynchronized (8.12%) first- and third-order troughs, resulting in the proportion of 4.07% and 4.78% of the total for the patterns S4 and D4.

In nonoverlapping area between swaths of two adjacent orbits, RMSE_r generally (90.6%) stays below 0.01, irrespective of rice-crop intensity. In swath overlapping area, RMSE_r exhibits a slight increase but mostly (87.8%) remains below 0.012. Since rice backscatter exhibits a substantial dynamic range (typically > 0.02) during the growing season, with the annual mean mostly (82.3%) exceeding 0.015, the harmonic fitting performance is considered acceptable. In other words, harmonic decomposition can effectively capture the periodic features of the original time series.

Research data are available to access: He, Ze; Li, Shihua (2023), "Mapping rice-crop intensity of Southern China in 2020," Mendeley Data, V4, doi: 10.17632/ktj22bsyhf.4.

REFERENCES

[1] J. Wei, Y. Cui, W. Luo, and Y. Luo, "Mapping paddy Rice distribution and cropping intensity in China from 2014 to 2019 with Landsat images, effective flood signals, and Google Earth engine," *Remote Sens.*, vol. 14, no. 3, p. 759, Feb. 2022.

[2] B. Qiu, W. Qi, Z. Tang, C. Chen, and X. Wang, "Rice cropping density and intensity lessened in Southeast China during the twenty-first century," *Environ. Monitor. Assessment*, vol. 188, no. 1, p. 5, Jan. 2016.

[3] G. Zhang et al., "Fingerprint of Rice paddies in spatial–temporal dynamics of atmospheric methane concentration in monsoon Asia," *Nature Commun.*, vol. 11, no. 1, p. 554, Jan. 2020.

[4] M. Boschetti et al., "PhenoRice: A method for automatic extraction of spatio-temporal information on Rice crops using satellite data time series," *Remote Sens. Environ.*, vol. 194, pp. 347–365, Jun. 2017.

[5] M. Jiang, L. Xin, X. Li, M. Tan, and R. Wang, "Decreasing Rice cropping intensity in Southern China from 1990 to 2015," *Remote Sens.*, vol. 11, no. 1, p. 35, Dec. 2018.

[6] X. Liu et al., "Annual dynamic dataset of global cropping intensity from 2001 to 2019," *Sci. Data*, vol. 8, no. 1, p. 283, Oct. 2021.

[7] B. Qiu et al., "Maps of cropping patterns in China during 2015–2021," *Sci. Data*, vol. 9, no. 1, p. 479, Aug. 2022.

[8] J. Han et al., "Annual paddy Rice planting area and cropping intensity datasets and their dynamics in the Asian monsoon region from 2000 to 2020," *Agricult. Syst.*, vol. 200, Jun. 2022, Art. no. 103437.

[9] H. V. T. Minh, R. Avtar, G. Mohan, P. Misra, and M. Kurasaki, "Monitoring and mapping of Rice cropping pattern in flooding area in the Vietnamese Mekong delta using Sentinel-1A data: A case of an Giang province," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 5, p. 211, May 2019.

[10] B. Minasny, R. M. Shah, N. C. Soh, C. Arif, and B. I. Setiawan, "Automated near-real-time mapping and monitoring of Rice extent, cropping patterns, and growth stages in Southeast Asia using Sentinel-1 time series on a Google Earth engine platform," *Remote Sens.*, vol. 11, no. 14, p. 1666, Jul. 2019.

[11] N. You and J. Dong, "Examining earliest identifiable timing of crops using all available Sentinel 1/2 imagery and Google Earth Engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 109–123, Mar. 2020.

[12] M. W. J. Davidson, T. Le Toan, F. Mattia, G. Satalino, T. Manninen, and M. Borgeaud, "On the characterization of agricultural soil roughness for radar remote sensing studies," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 2, pp. 630–640, Mar. 2000.

[13] K. Lasko, K. P. Vadrevu, V. T. Tran, and C. Justice, "Mapping double and single crop paddy Rice with Sentinel-1A at varying spatial scales and polarizations in Hanoi, Vietnam," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 2, pp. 498–512, Feb. 2018.

[14] P. Zhan, W. Zhu, and N. Li, "An automated Rice mapping method based on flooding signals in synthetic aperture radar time series," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112112.

[15] R. Fatchurrachman et al., "High-resolution mapping of paddy Rice extent and growth stages across peninsular Malaysia using a fusion of Sentinel-1 and 2 time series data in Google Earth engine," *Remote Sens.*, vol. 14, no. 8, p. 1875, Apr. 2022.

[16] Z. Yang, Y. Shao, K. Li, Q. Liu, L. Liu, and B. Brisco, "An improved scheme for Rice phenology estimation based on time-series multispectral HJ-1A/B and polarimetric RADARSAT-2 data," *Remote Sens. Environ.*, vol. 195, pp. 184–201, Jun. 2017.

[17] N. Torbick, D. Chowdhury, W. Salas, and J. Qi, "Monitoring Rice agriculture across Myanmar using time series Sentinel-1 assisted by Landsat-8 and PALSAR-2," *Remote Sens.*, vol. 9, no. 2, p. 119, Feb. 2017.

[18] D. B. Nguyen, A. Gruber, and W. Wagner, "Mapping Rice extent and cropping scheme in the Mekong delta using Sentinel-1A data," *Remote Sens. Lett.*, vol. 7, no. 12, pp. 1209–1218, Dec. 2016.

[19] M. Singha, J. Dong, G. Zhang, and X. Xiao, "High resolution paddy Rice maps in cloud-prone Bangladesh and northeast India using Sentinel-1 data," *Sci. Data*, vol. 6, no. 1, p. 26, Apr. 2019.

[20] D. Nguyen, K. Clauss, S. Cao, V. Naeimi, C. Kuenzer, and W. Wagner, "Mapping Rice seasonality in the Mekong delta with multi-year ENVISAT ASAR WSM data," *Remote Sens.*, vol. 7, no. 12, pp. 15868–15893, Nov. 2015.

[21] F. M. Bayer, D. M. Bayer, A. Marinoni, and P. Gamba, "A novel Rayleigh dynamical model for remote sensing data interpretation," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4989–4999, Jul. 2020.

[22] J. Tian et al., "Improving the accuracy of spring phenology detection by optimally smoothing satellite vegetation index time series based on local cloud frequency," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 29–44, Oct. 2021.

[23] Y. Chen, R. Cao, J. Chen, L. Liu, and B. Matsushita, "A practical approach to reconstruct high-quality Landsat NDVI time-series data by gap filling and the Savitzky–Golay filter," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 174–190, Oct. 2021.

[24] X. Li et al., "Phenology estimation of subtropical bamboo forests based on assimilated MODIS LAI time series data," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 262–277, Mar. 2021.

[25] J. Hrisko, P. Ramamurthy, Y. Yu, P. Yu, and D. Melecio-Vázquez, "Urban air temperature model using GOES-16 LST and a diurnal regressive neural network algorithm," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111495.

[26] M. Singha et al., "Identifying floods and flood-affected paddy Rice fields in Bangladesh based on Sentinel-1 imagery and Google Earth engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 278–293, Aug. 2020.

[27] D. Mandal, V. Kumar, A. Bhattacharya, Y. S. Rao, P. Siqueira, and S. Bera, "Sen4Rice: A processing chain for differentiating early and late transplanted Rice using time-series Sentinel-1 SAR data with Google Earth engine," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 12, pp. 1947–1951, Dec. 2018.

[28] B. A. Bradley, R. W. Jacob, J. F. Hermance, and J. F. Mustard, "A curve fitting procedure to derive inter-annual phenologies from time series of noisy satellite NDVI data," *Remote Sens. Environ.*, vol. 106, no. 2, pp. 137–145, Jan. 2007.

[29] S. Schlaffer, P. Matgen, M. Hollaus, and W. Wagner, "Flood detection from multi-temporal SAR data using harmonic analysis and change detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 38, pp. 15–24, Jun. 2015.

[30] S. Schlaffer, M. Chini, L. Giustarini, and P. Matgen, "Probabilistic mapping of flood-induced backscatter changes in SAR time series," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 56, pp. 77–87, Apr. 2017.

[31] M. E. Jakubauskas, D. R. Legates, and J. H. Kastens, "Crop identification using harmonic analysis of time-series AVHRR NDVI data," *Comput. Electron. Agricult.*, vol. 37, nos. 1–3, pp. 127–139, Dec. 2002.

[32] X. Zhang, B. Wu, G. Ponce-Campos, M. Zhang, S. Chang, and F. Tian, "Mapping up-to-date paddy Rice extent at 10 m resolution in China through the integration of optical and synthetic aperture radar images," *Remote Sens.*, vol. 10, no. 8, p. 1200, Jul. 2018.

[33] Y. He et al., "Examining Rice distribution and cropping intensity in a mixed single- and double-cropping region in South China using all available sentinel 1/2 images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 101, Sep. 2021, Art. no. 102351.

[34] H. K. Hoang, M. Bernier, S. Duchesne, and Y. M. Tran, "Rice mapping using RADARSAT-2 dual- and quad-pol data in a complex land-use Watershed: Cau River Basin (Vietnam)," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 7, pp. 3082–3096, Jul. 2016.

[35] J. Duan and G. Zhou, "Climatic suitability of double Rice planting regions in China," *Scientia Agricultura Sinica*, vol. 45, no. 2, pp. 218–227, Oct. 2012.

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[36] Q. Xu et al., "Effects of low temperature on the growth and development of Rice plants and the advance of regulation pathways: A review," *Chin. J. Rice Sci.*, vol. 36, no. 2, pp. 118–130, Nov. 2022.
[37] G. Yang, H. Shen, L. Zhang, Z. He, and X. Li, "A moving weighted harmonic analysis method for reconstructing high-quality SPOT VEGETATION NDVI time-series data," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6008–6021, Nov. 2015.
[38] V. Louf, O. Pujol, H. Sauvageot, and J. Riédi, "Simulation of airborne radar observations of precipitating systems at various frequency bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 3, pp. 1627–1634, Mar. 2014.
[39] B. A. Chuma et al., "Altitude, temperature, and n management effects on yield and yield components of contrasting lowland Rice cultivars," *J. Agronomy Crop Sci.*, vol. 206, no. 4, pp. 456–465, Aug. 2020.
[40] C. Zhang et al., "Study on the influence of composite soil on the slope stability of farmland during in land consolidation," *Environ. Earth Sci.*, vol. 81, no. 5, p. 166, Mar. 2022.
[41] R. Shen et al., "High-resolution distribution maps of single-season Rice in China from 2017 to 2022," *Earth Syst. Sci. Data*, vol. 15, no. 7, pp. 3203–3222, Jul. 2023.
[42] B. Pan et al., "High resolution distribution dataset of double-season paddy Rice in China," *Remote Sens.*, vol. 13, no. 22, p. 4609, Nov. 2021.
[43] S. Xu et al., "A robust index to extract paddy fields in cloudy regions from SAR time series," *Remote Sens. Environ.*, vol. 285, Feb. 2023, Art. no. 113374.



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